

Module 21: Metrology and Precision Measurement

Module 21 - Metrology and Precision Measurement

21.1.1 The Role of Measurement in Precision Machining

Metrology is the science of measurement, and in CNC machining, it serves as the foundation for quality assurance, process control, and continuous improvement. Without accurate and reliable measurement, it is impossible to verify that machined parts meet specifications, diagnose machining problems, or optimize processes.

The Measurement-Manufacturing Loop

In professional CNC operations, measurement is not an afterthought—it is integrated throughout the entire manufacturing cycle:

1. **Pre-Production:** Calibration of machine tools, verification of fixtures and tooling
2. **Setup:** Workpiece datum establishment, tool length measurement, alignment verification
3. **In-Process:** On-machine probing, adaptive control, tool wear monitoring
4. **Post-Production:** Final inspection, documentation, process feedback

Why Metrology Matters

- **Quality Assurance:** Verify parts meet design specifications and customer requirements
 - **Process Control:** Monitor machining processes and detect trends before defects occur
 - **Cost Reduction:** Catch errors early, reduce scrap and rework
 - **Legal Compliance:** Meet industry standards (aerospace AS9100, medical ISO 13485)
 - **Continuous Improvement:** Data-driven optimization of machining parameters
 - **Machine Maintenance:** Track machine accuracy over time, predict maintenance needs
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21.1.2 Fundamental Measurement Principles

21.1.2.1 Accuracy vs. Precision

These terms are often confused, but they represent distinct concepts:

Accuracy refers to how close a measurement is to the true value. An accurate measurement has minimal systematic error (bias).

Precision refers to the repeatability of measurements. Precise measurements cluster tightly together, even if they are not accurate.

Analogy: Think of a target: - High accuracy, high precision: All shots hit the bullseye - High accuracy, low precision: Shots scatter around the bullseye - Low accuracy, high precision: Shots cluster together but away from bullseye - Low accuracy, low precision: Shots scatter everywhere

In Practice: - A micrometer consistently reading 0.005" high has good precision but poor accuracy - A worn caliper giving random readings has poor precision (regardless of accuracy) - The goal is always both: accurate AND precise measurements

21.1.2.2 Resolution and Repeatability

Resolution is the smallest increment that a measuring instrument can detect and display.

- A digital caliper with 0.0005" (0.01mm) resolution cannot reliably measure differences smaller than 0.0005"
- Higher resolution != higher accuracy (a fine-resolution instrument can still be inaccurate)
- Rule of thumb: Instrument resolution should be $\leq 10\%$ of the tolerance being measured

Repeatability is the variation in measurements when the same person measures the same feature multiple times under identical conditions.

- Good repeatability indicates a stable measurement process
- Poor repeatability suggests problems with technique, instrument, or environment
- Repeatability is assessed using standard deviation (sigma) of repeated measurements

Reproducibility extends this concept: Can different operators achieve similar results?

21.1.2.3 Traceability and Standards

Metrological Traceability is an unbroken chain of calibrations linking a measurement to a recognized standard (typically maintained by a national metrology institute like NIST in the USA).

The Traceability Chain: 1. **International Standard:** SI units defined by international agreement (meter, kilogram, second, etc.) 2. **National Standard:** Maintained by national labs (NIST, NPL, PTB) 3. **Reference Standard:** Used by calibration laboratories 4. **Working Standard:** Used to calibrate shop instruments 5. **Shop Instruments:** Used for daily production measurements

Why Traceability Matters: - Ensures measurements are consistent worldwide - Required for ISO certification and customer audits - Provides legal defensibility of inspection results - Enables comparison of measurements across different facilities

Calibration Certificates document the traceability chain and provide: - Measured values vs. reference standards - Measurement uncertainty - Date of calibration and due date for next calibration - Environmental conditions during calibration

21.1.3 Sources of Measurement Error

All measurements contain error. Understanding error sources is essential for controlling them and estimating measurement uncertainty.

21.1.3.1 Systematic Errors

Systematic errors are repeatable, predictable errors that bias measurements in a consistent direction.

Common Systematic Errors:

1. **Calibration Error:** Instrument is not properly calibrated to standards
 - Example: Micrometer reads 0.003" when closed (zero error)
 - Solution: Calibrate instrument, apply correction factor
2. **Thermal Expansion:** Parts and instruments expand/contract with temperature
 - Standard reference temperature: 20°C (68°F)
 - Aluminum expands 23 ppm/°C; steel 12 ppm/°C
 - Example: A 300mm aluminum part at 25°C is 0.035mm longer than at 20°C

- Solution: Temperature-controlled environment, thermal compensation
3. **Cosine Error:** Measuring at an angle instead of perpendicular to surface
 - Measured value = True value $\times \cos(\theta)$
 - Even small angles introduce significant error
 - Solution: Ensure perpendicular alignment, use fixtures
 4. **Abbe Error:** Measurement axis does not coincide with the axis of interest
 - Error = Offset distance $\times$ Angular error
 - Common in machines with non-coaxial scales
 - Solution: Minimize offset, use Abbe-compliant designs
 5. **Contact Force Variation:** Inconsistent pressure on measurement probes
 - Hard contact deforms soft materials (aluminum, plastics)
 - Light contact on rough surfaces gives erratic readings
 - Solution: Use consistent technique, controlled contact force

21.1.3.2 Random Errors

Random errors are unpredictable variations that cause measurements to scatter around the true value.

Sources of Random Error:

1. **Operator Variability:** Differences in technique, feel, alignment
2. **Instrument Noise:** Electronic noise in digital instruments
3. **Surface Finish:** Peaks and valleys on rough surfaces
4. **Vibration:** External vibrations affecting measurement stability
5. **Air Currents:** Thermal gradients causing drift in sensitive instruments

Dealing with Random Errors: - Take multiple measurements and average them - Use statistical methods (standard deviation, confidence intervals) - Reduce environmental disturbances - Improve operator training and technique

21.1.3.3 Environmental Factors

The measurement environment has enormous impact on accuracy:

Temperature: - Most significant environmental factor in precision measurement - Metal parts expand/contract approximately 10-25 ppm/°C - Solution: Climate-controlled room (20°C $\pm$ 1°C), thermal soaking, correction factors

Humidity: - Affects dimensional stability of non-metals (plastics, composites) - Can cause corrosion on precision surfaces - Solution: Control humidity (typically 45-55% RH)

Vibration: - Affects sensitive instruments (CMMs, laser systems) - Sources: nearby machinery, traffic, HVAC systems - Solution: Isolated foundation, vibration damping mounts

Air Cleanliness: - Dust and contamination on precision surfaces causes false readings - Particulates in air can settle on gages and parts - Solution: Clean room or filtered air, part cleaning procedures

Lighting: - Critical for visual inspection and optical measurement - Glare and shadows affect operator judgments - Solution: Appropriate task lighting, eliminate reflections

Measurement Uncertainty

Every measurement has uncertainty—the range within which the true value is likely to lie.

Uncertainty Budget accounts for all error sources: - Instrument calibration uncertainty - Resolution uncertainty - Repeatability uncertainty - Environmental uncertainty - Operator variability

Reporting Uncertainty: Measurements should be reported as: *Measured Value +/- Uncertainty* (e.g., 25.347 mm +/- 0.003 mm)

The 10:1 Rule: For inspection to be reliable, the measurement uncertainty should be $\leq 10\%$ of the tolerance being verified.

Example: To measure a dimension with $\pm 0.010''$ tolerance, measurement uncertainty should be $\leq 0.001''$

Summary

Metrology provides the foundation for quality manufacturing. Understanding the fundamental principles—accuracy, precision, resolution, repeatability, traceability—and the sources of measurement error enables proper selection and use of measurement instruments.

In the next section, we'll explore measurement standards and calibration systems that ensure measurements are accurate and traceable.

Key Takeaways

1. **Measurement is integral** to every phase of CNC manufacturing
 2. **Accuracy and precision** are distinct concepts; both are necessary
 3. **Resolution** must be appropriate for the tolerance being measured
 4. **Traceability** links measurements to international standards
 5. **Systematic errors** are repeatable and can be corrected
 6. **Random errors** require statistical methods to control
 7. **Environmental control** is critical for precision measurement
 8. **Measurement uncertainty** must be quantified and reported
 9. **The 10:1 rule** ensures adequate measurement capability
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Review Questions

1. What is the difference between accuracy and precision? Can a measurement be precise but not accurate?
2. Why is metrological traceability important in regulated industries like aerospace?
3. Calculate the thermal expansion of a 500mm steel part when temperature increases from 20°C to 25°C (steel $\alpha = 12 \text{ ppm}/^\circ\text{C}$)
4. What is Abbe error and how can it be minimized in machine tool design?
5. If a tolerance is $\pm 0.005''$, what maximum measurement uncertainty is acceptable per the 10:1 rule?

Module 21 - Metrology and Precision Measurement

Effective inspection planning ensures parts meet specifications efficiently while maintaining traceability and documentation for quality management systems.

21.10.1 First Article Inspection (FAI)

First Article Inspection is a comprehensive inspection of the first production part(s) to verify the manufacturing process produces parts meeting all design requirements.

Purpose of FAI

Verify: - Manufacturing process capability - Tooling and fixturing adequacy - Program accuracy - Drawing interpretation - Supplier understanding of requirements

When Required: - New part introduction - After engineering change - Process change (new machine, method, location) - After significant production break - Customer or contract requirement

FAI Standards

AS9102: Aerospace First Article Inspection Requirement - Defines documentation and procedures - Form 1: Part number accountability - Form 2: Product accountability (characteristic accountability) - Form 3: Characteristic accountability and verification

Industry Specific: Automotive (PPAP), Medical (Design History File), etc.

FAI Process

Step 1: Planning - Identify all characteristics on drawing - Determine measurement method for each feature - Establish datum reference frames - Select appropriate instruments

Step 2: Measurement - Measure all characteristics - Record actual values (not just pass/fail) - Include measurement uncertainty where applicable - Photograph or scan critical features

Step 3: Documentation - Complete forms (AS9102 or equivalent) - Record measurements vs. specifications - Note any deviations - Reference calibrated instruments used

Step 4: Review and Approval - Engineering review - Quality review - Customer approval (if required)

Step 5: Retention - Maintain records per customer/industry requirements - Typically 7-10 years or life of program

FAI Documentation

Required Information: - Part number, revision, quantity - Customer name and purchase order - All dimensional characteristics - Material verification - Special processes (heat treat, plating, etc.) - Inspection equipment used - Inspector and date - Approval signatures

Supporting Documents: - Engineering drawing - Material certifications - Process certifications (heat treat, NDT, etc.) - Instrument calibration records - Photographs or comparison to master

21.10.2 Production Part Approval Process (PPAP)

PPAP is the automotive industry standard for part approval (defined by AIAG - Automotive Industry Action Group).

PPAP Purpose

Demonstrates that: - Supplier understands customer requirements - Process can consistently produce conforming parts - Actual production process, tooling, and equipment will be used

PPAP Levels

Level 1: Part Submission Warrant (PSW) only - Lowest level, minimal documentation

Level 2: PSW with product samples and limited supporting data

Level 3: PSW with product samples and complete supporting data - Most common level

Level 4: PSW and complete supporting data for review at supplier's site

Level 5: PSW and complete supporting data reviewed at customer's site - Highest level, typically for critical parts

PPAP Requirements (Level 3)

18 Elements:

1. Design Records (if supplier responsible for design)
2. Engineering Change Documents (authorization)
3. Customer Engineering Approval (if required)
4. Design FMEA (if supplier responsible)
5. Process Flow Diagram
6. Process FMEA
7. Control Plan
8. Measurement System Analysis (MSA/Gage R&R)
9. Dimensional Results (inspection data)
10. Material/Performance Test Results
11. Initial Process Studies (capability studies)
12. Qualified Laboratory Documentation
13. Appearance Approval Report (AAR) if applicable
14. Sample Product (physical samples)
15. Master Sample (retained by supplier)
16. Checking Aids (fixtures, gages)
17. Customer-Specific Requirements
18. Part Submission Warrant (PSW)

Control Plan

The Control Plan is a living document describing how quality will be controlled during production.

Content: - Part number and description - Process steps - Characteristics to be controlled at each step - Methods and specifications - Sample size and frequency - Control method (SPC, 100% inspection, etc.) - Reaction plan for out-of-spec conditions

Types: - **Prototype:** During development - **Pre-Launch:** Before full production - **Production:** Ongoing manufacturing

Initial Process Capability Studies

Required for special characteristics:

Data Collection: - Minimum 25-30 subgroups - Subgroup size typically 3-5 parts - Consecutive production - Stable process conditions

Analysis: - Calculate Cpk - Typical minimum requirement: $Cpk \geq 1.33$ - Critical characteristics: $Cpk \geq 1.67$

Actions if Insufficient Capability: - 100% inspection until improved - Increase control plan monitoring
- Document containment plan

21.10.3 Inspection Planning from Drawings

Effective inspection planning translates drawing requirements into measurement procedures.

Analyzing the Drawing

Step 1: Identify All Features - Dimensions and tolerances - GD&T callouts - Surface finish - Material specifications - Notes and general tolerances

Step 2: Classify Characteristics

Critical: Affects safety or regulatory compliance - 100% inspection or very frequent sampling - Tightest controls

Major: Affects fit, form, function - SPC monitoring or periodic sampling - Standard controls

Minor: Non-functional, appearance - Periodic sampling - Reduced controls

Step 3: Establish Datum Reference Frames - Identify datum features in order (A, B, C) - Determine how to simulate datums physically - Plan fixture or measurement sequence

Step 4: Select Measurement Methods

Consider: - Tolerance magnitude (10:1 rule) - Feature type (hole, slot, profile, etc.) - Access (internal, external, small, large) - Production volume (dedicated gage vs. CMM) - Cost and cycle time

Inspection Documentation

Inspection Plan includes:

For Each Characteristic: 1. Feature description 2. Specification and tolerance 3. Measurement method 4. Instrument required (with ID number) 5. Sample size and frequency 6. Acceptance criteria 7. Reaction plan if out of spec

Example Entry:

Feature: Hole $\varnothing 10.00 +0.05/-0.00$

Method: Dial bore gage

Instrument: DBG-001 (cal due 6/2025)

Frequency: First piece, last piece, every 25th piece

Accept: 9.95-10.05 mm

React: Stop production, notify supervisor, 100% sort

Measurement Sequence

Logical Order: 1. Establish primary datum (typically on surface plate) 2. Establish secondary and tertiary datums 3. Measure features in order that maintains datum reference 4. Avoid re-handling that loses datum reference

Efficiency: - Group measurements by instrument - Group by setup or orientation - Minimize part handling
- Consider dedicated fixtures for high volume

21.10.4 Measuring Uncertainty and Reporting

Measurement uncertainty quantifies the doubt in a measurement result.

Sources of Uncertainty

Type A (Statistical): - Repeatability of instrument - Reproducibility between operators - Environmental variation

Type B (Other sources): - Calibration uncertainty - Resolution uncertainty - Temperature effects - Instrument drift - Fixture errors

Calculating Combined Uncertainty

Simplified Approach:

$$U_{\text{combined}} = \sqrt{(u_1^2 + u_2^2 + u_3^2 + \dots + u_n^2)}$$

Where u_1 , u_2 , etc. are individual uncertainty components

Example: - Calibration uncertainty: ± 0.002 mm - Repeatability (from Gage R&R): ± 0.003 mm - Temperature effect: ± 0.001 mm

$$U_{\text{combined}} = \sqrt{(0.002^2 + 0.003^2 + 0.001^2)}$$

$$U_{\text{combined}} = \sqrt{(0.000004 + 0.000009 + 0.000001)}$$

$$U_{\text{combined}} = \sqrt{0.000014} = 0.0037 \text{ mm}$$

Expanded Uncertainty (95% confidence):

$$U = k \times U_{\text{combined}}$$

Where $k = 2$ for 95% confidence

Example: $U = 2 \times 0.0037 = 0.007$ mm

Reporting Measurements

Proper Format:

Measured value: 25.347 mm ± 0.007 mm ($k=2$)

Or:

Measured value: 25.347 mm

Measurement uncertainty: ± 0.007 mm (95% confidence)

Decision Rules (ISO 14253-1)

When measurement uncertainty is significant relative to tolerance:

Simple Acceptance: Accept if measured value is within specification limits - Risk of accepting parts that are actually out of spec

Strict Acceptance (Guard Banding): Accept only if measured value is within reduced limits - Reduced limits = Spec limits minus uncertainty - Conservative but reduces risk

Example: - Specification: 25.00 ± 0.05 mm (24.95-25.05) - Measurement uncertainty: ± 0.01 mm - Guard band limits: 24.96-25.04 mm

21.10.5 Quality Management Systems (ISO 9001)

ISO 9001 is the international standard for quality management systems.

ISO 9001 Key Elements

Customer Focus: Understand and meet customer requirements

Leadership: Top management commitment to quality

Process Approach: Manage activities as processes

Improvement: Continual improvement of QMS

Evidence-Based Decision Making: Use data and analysis

Relevant Clauses for Metrology

Clause 7.1.5 - Monitoring and Measuring Resources: - Identify required measurement equipment - Ensure equipment is suitable for purpose - Maintain and calibrate as needed - Safeguard from damage and deterioration - Verify before use if removed from controlled area

Calibration Requirements: - Calibrated or verified at specified intervals - Traceable to international or national standards - Identified calibration status - Safeguarded from adjustments that invalidate results - Records of calibration maintained

Clause 8.6 - Release of Products: - Verify product meets acceptance criteria - Maintain evidence of conformity - Traceability to person(s) authorizing release

Clause 10 - Improvement: - Nonconformity and corrective action - Continual improvement

Metrology System Requirements

Documentation: - Calibration procedures - Measurement procedures - Instrument lists and recall system - Calibration records - Uncertainty analysis (where required)

Controls: - Calibration interval management - Out-of-tolerance procedures - Measurement uncertainty consideration - Environmental controls - Handling and storage

Training: - Qualified personnel for measurements - Documented competency - Ongoing training records

21.10.6 Documentation and Record Keeping

Proper documentation provides traceability, enables process improvement, and supports quality system requirements.

Types of Records

Instrument Records: - Instrument inventory list - Calibration certificates - Calibration history - Repair and maintenance records - Gage R&R studies

Inspection Records: - First article inspection reports - In-process inspection data - Final inspection reports - SPC charts - Capability studies

Nonconformance Records: - Nonconforming material reports - Corrective action requests - Customer complaints - Scrap and rework records

Audit Records: - Internal audit reports - Customer audit findings - Corrective actions and effectiveness

Electronic Record Systems

Quality Management Software (QMS): - Centralized database - Automated workflows - Revision control - Electronic signatures - Reporting and analysis

Calibration Management Software: - Instrument database - Automated recall notifications - Calibration history tracking - Certificate repository - Statistical analysis

Benefits: - Improved traceability - Reduced paperwork - Faster retrieval - Better analysis capability - Audit readiness

Record Retention

Typical Requirements: - General quality records: 3-10 years - Calibration records: Life of instrument + 1 year - PPAP/FAI records: Life of program - Medical devices: Often lifetime of device + years - Aerospace: Often indefinite

Storage Requirements: - Secure and backed up - Protected from damage - Accessible for audits - Controlled access - Version control for documents

Summary

Inspection planning transforms drawing requirements into actionable measurement procedures. First Article Inspection verifies new or changed processes. PPAP provides comprehensive part approval for automotive suppliers. Proper uncertainty analysis ensures measurement reliability. Quality management systems (ISO 9001) provide the framework for effective metrology operations. Complete documentation enables traceability and continuous improvement.

Key Takeaways

1. **FAI** verifies manufacturing process capability for new or changed parts
2. **AS9102** defines aerospace FAI requirements
3. **PPAP** is the automotive part approval standard (AIAG)
4. **Control Plan** describes how quality will be maintained in production
5. **Process capability** ($C_{pk} \geq 1.33$) is typically required for PPAP
6. **Inspection planning** must consider datum reference frames per GD&T
7. **Measurement uncertainty** quantifies confidence in results
8. **Guard banding** reduces risk of accepting out-of-spec parts
9. **ISO 9001** requires calibration traceability and management
10. **Documentation** provides traceability and supports improvement

Review Questions

1. What is the purpose of First Article Inspection and when is it required?
2. List five of the 18 PPAP requirements.
3. What is a Control Plan and what information does it contain?
4. What is the typical minimum C_{pk} requirement for PPAP approval?
5. How do you calculate combined measurement uncertainty from multiple sources?
6. What is guard banding and when should it be used?
7. What are the ISO 9001 requirements for calibration of measurement equipment?
8. Why is proper record retention important in regulated industries?

Module 21 - Metrology and Precision Measurement

Advanced metrology technologies enable measurement of complex geometries, internal features, and microscopic details that traditional methods cannot address.

21.11.1 3D Scanning and Reverse Engineering

3D scanning captures the complete surface geometry of parts, enabling comparison to CAD, reverse engineering, and inspection of complex freeform surfaces.

3D Scanning Technologies

Structured Light Scanning:

Principle: - Project patterns of light onto object - Cameras capture distorted patterns - Triangulation calculates 3D coordinates - Millions of points captured in seconds

Characteristics: - Very fast data acquisition - High resolution (10-100 μm typical) - Non-contact - Portable systems available - Sensitive to lighting and surface finish

Applications: - Reverse engineering - Large part inspection - Deformation analysis - Art and heritage preservation

Laser Line Scanning:

Principle: - Laser line projected on surface - Camera images line from angle - Triangulation determines profile - Scan head moves to build 3D surface

Characteristics: - Accurate (20-50 μm typical) - Works on variety of surfaces - Can be integrated with CMM or robotic arm - Good for complex shapes

Applications: - Turbine blade inspection - Body panel measurement - Castings and forgings - Tool and die verification

Photogrammetry:

Principle: - Multiple photographs from different angles - Software identifies common points - Calculates 3D coordinates - Requires coded or natural targets

Characteristics: - Portable (camera-based) - Scales to very large objects (meters to tens of meters) - Accuracy: 10-100 μm depending on setup - Requires overlapping images

Applications: - Large assemblies - Aircraft and vehicles - Shipbuilding - On-site measurement

Point Cloud Processing

Point Cloud: Collection of millions of XYZ coordinates representing surface

Processing Steps:

1. **Alignment:** Register multiple scans to common coordinate system
2. **Filtering:** Remove noise and outliers
3. **Meshing:** Create triangulated surface (STL mesh)
4. **Simplification:** Reduce point count while maintaining detail
5. **Comparison:** Analyze deviation from CAD model

CAD Comparison: - Import CAD model - Best-fit align point cloud to CAD - Generate color deviation map - Quantify dimensional differences - Identify areas out of tolerance

Color Maps: Visual representation of deviations - Blue: Negative deviation (part smaller than nominal) - Green: Within tolerance - Red: Positive deviation (part larger than nominal)

Reverse Engineering

Process:

1. **Scan:** Capture complete part geometry
2. **Process:** Clean and optimize point cloud
3. **Surface Creation:** Fit surfaces to point cloud
 - Parametric surfaces for regular features
 - NURBS or subdivision surfaces for freeform
4. **Feature Recognition:** Identify holes, fillets, planes
5. **CAD Model:** Create solid model from surfaces
6. **Verification:** Compare CAD to original scan

Applications: - Recreate obsolete parts without drawings - Digitize clay models or handmade masters - Duplicate organic or sculptured forms - Create CAD from physical prototypes

Challenges: - No design intent (dimensions, relationships) - Time-consuming for complex parts - Requires skilled CAD modeler - Difficult to capture sharp edges and fine features

21.11.2 X-Ray and CT Scanning for Internal Features

Computed Tomography (CT) enables non-destructive inspection of internal features impossible to measure by other methods.

Industrial CT Scanning

Principle: - X-rays pass through part from multiple angles - Detector measures attenuation - Computer reconstructs 3D volume - Creates “virtual cross-sections”

Capabilities:

Internal Geometry: - Hidden cavities and voids - Internal channels and passages - Wall thickness - Porosity and defects

Assembly Analysis: - Component positions - Gap and flush measurements - Interference detection - Weld and bond inspection

Material Analysis: - Density variations - Material identification - Inclusions and contamination

CT Measurement Accuracy

Resolution: 5-100 μm depending on part size and system

Accuracy Factors: - Part size vs. scanning volume - Material density and thickness - Beam energy and quality - Reconstruction algorithms - Surface determination method

Typical Accuracy: 10-50 μm for precision systems

Limitations: - Dense materials (steel) require high-energy systems - Large parts more difficult than small - Slow scan time (minutes to hours) - Expensive equipment - Radiation safety requirements

Applications

Aerospace: - Turbine blade internal cooling passages - Additive manufacturing inspection - Composite layup verification - Weld inspection

Medical Devices: - Implant metrology - Polymer component inspection - Assembly verification

Automotive: - Powertrain components - Injection-molded plastic parts - Electronic assemblies

Additive Manufacturing: - Porosity analysis - Internal lattice structures - Residual powder detection - Critical for AS9100 qualification

First Article Inspection: - Complete dimensional verification including internals - Virtual assembly checks
- Non-destructive evaluation

21.11.3 Interferometric Microscopy

Interferometric microscopy combines microscope optics with interferometry for sub-nanometer resolution surface measurement.

White Light Interferometry (WLI)

Principle: - Broadband white light through microscope objective - Beam splitter creates reference and measurement beams - Recombine to create interference fringes - Vertical scan creates 3D surface map

Capabilities: - Vertical resolution: Sub-nanometer (0.1 nm) - Lateral resolution: 0.5-10 μm (depends on objective) - Field of view: 100 μm to several mm - 3D surface measurement

Applications: - Ultra-smooth surface measurement (optical surfaces, wafers) - Micro-scale features (MEMS, micro-machined parts) - Surface roughness ($R_a < 1 \text{ nm}$ to several μm) - Step height measurement - Thin film thickness

Advantages: - Non-contact - Very high resolution - Fast measurement (seconds) - 3D surface maps

Limitations: - Small field of view - Requires flat, mostly reflective surfaces - Cannot measure steep slopes ($>20\text{-}30^\circ$) - Expensive equipment

Phase-Shifting Interferometry (PSI)

Principle: - Monochromatic laser light - Reference mirror shifts in controlled steps - Multiple images captured - Phase calculation determines height

Characteristics: - Sub-nanometer vertical resolution - Higher precision than WLI - Requires very smooth surfaces - Limited to surfaces within one wavelength variation

Applications: - Optical components (lenses, mirrors) - Semiconductor wafers - Precision machined surfaces - Gage block flatness

Confocal Microscopy

Principle: - Only in-focus light passes pinhole to detector - Scanning in Z creates 3D image - Excellent depth discrimination

Characteristics: - High lateral and vertical resolution - Can handle steep slopes - Slower than interferometry - Works on variety of surfaces

Applications: - Rough surfaces - Complex micro-geometry - Multi-layer structures - Materials analysis

21.11.4 Form Measurement of Freeform Surfaces

Freeform surfaces (sculptured, organic, or mathematically complex) present unique measurement challenges.

Defining Freeform Surfaces

Freeform: Surfaces that cannot be described by simple geometric primitives (planes, cylinders, spheres, cones)

Examples: - Turbine blade airfoils - Injection mold cavities - Artistic sculptures - Medical implants - Optical aspheres - Consumer product housings

Measurement Methods

CMM with Scanning Probe:

Process: 1. Import CAD model 2. Generate scan path along surface 3. Probe continuously contacts and scans surface 4. Thousands of points captured 5. Best-fit alignment to CAD 6. Deviation analysis

Capabilities: - High accuracy (2-10 μm) - Large parts (up to machine envelope) - Tactile measurement (actual surface)

Limitations: - Requires CAD model for programming - Time-consuming (minutes per feature) - Limited by probe access

Optical Scanning (Laser, Structured Light):

Process: 1. Scan surface from multiple orientations 2. Align scans to common reference 3. Create point cloud or mesh 4. Compare to CAD

Advantages: - Very fast (seconds to minutes) - Non-contact - Complete surface capture

Limitations: - Surface finish affects accuracy - Reflective or transparent surfaces problematic - Lower accuracy than CMM (20-100 μm typical)

Form Deviation Analysis

Best-Fit Alignment: - Minimize RMS deviation between measured and nominal - Removes position and orientation errors - Focuses on form errors

Deviation Metrics: - **Peak positive deviation:** Maximum material condition - **Peak negative deviation:** Minimum material condition - **RMS deviation:** Overall average error - **Standard deviation:** Variation in form

Color Deviation Maps: - Visual representation of form errors - Quickly identify problem areas - Communicate results effectively

Profile Tolerancing

GD&T Profile of a Surface: - Specifies allowable form deviation - With datums: Controls form, orientation, and location - Without datums: Controls form only

Inspection Strategy: 1. Establish datum reference frame (if applicable) 2. Measure surface (scan or probe) 3. Best-fit to nominal (if no datums) or align to datums 4. Calculate maximum deviation 5. Verify all points within tolerance zone

21.11.5 Measurement in Non-Standard Environments

Precision measurement traditionally requires controlled environments, but field measurement and in-process measurement require techniques for challenging conditions.

Portable Metrology

Laser Trackers:

Capabilities: - Measure 3D coordinates up to 80 meters range - Accuracy: 10-50 μm + distance-dependent error - Portable and quickly relocatable - Follows retroreflector target

Applications: - Large part assembly (aircraft, ships) - Tooling alignment - Machine tool calibration - On-site inspection

Articulated Arm CMMs:

Capabilities: - 6 or 7-axis portable arm - Typical range: 2-4 meters - Accuracy: 25-80 μm - Bring measurement to part

Applications: - Field inspection - Large assemblies - Reverse engineering - Parts that cannot move to CMM

Photogrammetry Systems:

Capabilities: - Camera-based measurement - Scales to very large objects - Portable and flexible

Applications: - Aircraft positioning - Shipbuilding - Wind turbine blades - On-site dimensional control

Environmental Compensation

Temperature Compensation: - Measure part and ambient temperature - Apply material CTE corrections - Reference to 20°C standard

Strategies: - Allow thermal soak time - Shield from direct sunlight - Measure early morning (stable temp) - Apply software compensation

Vibration Isolation: - Measure during quiet periods - Increase measurement time (averaging) - Isolate instrument from vibration sources

Humidity Control: - Critical for hygroscopic materials - Seal and condition parts before measuring - Environmental data logger

In-Situ Measurement

On-Machine Measurement: - Covered in Section 21.5 - Measurement during manufacturing - Accepts less-than-ideal environment

Process-Integrated Sensors: - Laser micrometers in production lines - Vision systems for inline inspection - Automated dimensional checks

Challenges: - Coolant, chips, and contamination - Vibration and thermal effects - Speed vs. accuracy tradeoffs

Summary

Advanced metrology extends measurement capability to previously impossible applications. 3D scanning captures complete surface geometry for inspection and reverse engineering. CT scanning reveals internal features non-destructively. Interferometric microscopy achieves sub-nanometer resolution. Freeform surface measurement enables complex part inspection. Portable metrology brings precision to the field. These advanced technologies complement traditional methods and expand the scope of measurable features.

Key Takeaways

1. **3D scanning** captures millions of points for complete surface inspection
 2. **Structured light and laser scanning** are fastest methods for complex surfaces
 3. **CT scanning** enables non-destructive internal inspection
 4. **Additive manufacturing** heavily relies on CT for qualification
 5. **Interferometric microscopy** achieves sub-nanometer resolution
 6. **Freeform surfaces** require CAD comparison and profile tolerancing
 7. **Portable metrology** (laser trackers, arms) enables field measurement
 8. **Environmental compensation** extends measurement to non-ideal conditions
 9. **Point cloud processing** transforms scan data into actionable information
 10. **Advanced metrology** complements, not replaces, traditional methods
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Review Questions

1. What are the main differences between structured light scanning and laser line scanning?
2. What types of features can CT scanning measure that traditional CMMs cannot?
3. What is the typical vertical resolution achievable with white light interferometry?
4. Describe the process of reverse engineering a part using 3D scanning.
5. What is a color deviation map and how is it used?
6. What are the advantages of laser trackers over traditional CMMs for large part measurement?
7. Why is CT scanning particularly important for additive manufacturing inspection?
8. What factors affect the accuracy of industrial CT scanning?

Module 21 - Metrology and Precision Measurement

This section presents real-world metrology applications across industries, demonstrating how measurement principles and techniques solve practical manufacturing challenges.

21.12.1 Metrology for Aerospace Components

Aerospace manufacturing demands the highest level of precision and documentation due to safety-critical applications.

Case Study: Turbine Blade Inspection

Component: Gas turbine blade for jet engine **Material:** Nickel superalloy (Inconel 718) **Challenges:** Complex freeform airfoil, tight tolerances, internal cooling passages

Inspection Requirements:

1. **Airfoil Profile** (GD&T Profile of a Surface)
 - Tolerance: 0.010" (0.25mm)
 - Critical for aerodynamic performance
 - Datum reference: Root attachment face and edges
2. **Root Attachment** (Fir-tree geometry)
 - Position tolerance: 0.002" (0.05mm)
 - Critical for load transfer and safety
3. **Internal Cooling Passages**
 - Non-visible, must be verified non-destructively
 - Diameter: 0.040-0.080" (1-2mm)
 - Position and connectivity critical
4. **Surface Finish**
 - Airfoil: Ra 32 μ m (0.8 μ m)
 - Root: Ra 63 μ m (1.6 μ m)
 - Affects fatigue life and efficiency

Measurement Solution:

CMM with Scanning Probe: - Import CAD model of airfoil - Generate scan path (500-1000 points) - Establish datum reference frame from root features - Scan airfoil surface continuously - Best-fit alignment and deviation analysis - Generate color deviation map - Typical inspection time: 15-20 minutes

CT Scanning (for internal passages): - Non-destructive inspection of cooling holes - Verify diameter, position, and connectivity - Detect blockages or incomplete drilling - 3D visualization for analysis - Required for first article and periodic validation - Scan time: 1-2 hours

Surface Profilometry: - Stylus measurement at specified locations - Multiple measurements per zone - Average Ra reported - Spot check during production

Results: - First article inspection fully documented per AS9102 - Production parts verified with sampling plan - CMM data feeds SPC for process monitoring - CT scans archived for traceability

Outcome: - Zero engine failures due to blade defects - Process Cpk > 1.67 for critical features - Customer approval with full traceability

21.12.2 Medical Device Inspection Requirements

Medical devices require stringent inspection and documentation per FDA regulations and ISO 13485.

Case Study: Orthopedic Hip Implant

Component: Cobalt-chromium femoral stem **Application:** Total hip replacement **Regulatory:** Class III medical device (high risk)

Critical Features:

1. **Taper Morse Taper** ($5^{\circ}43'$)
 - Tolerance: $\pm 0.0005''$ ($\pm 0.013\text{mm}$) on diameter
 - Angle tolerance: ± 5 arcminutes
 - Surface finish: Ra 8 μin ($0.2\text{ }\mu\text{m}$)
 - Critical for modular assembly
2. **Stem Geometry**
 - Profile tolerance: $\pm 0.005''$ ($\pm 0.13\text{mm}$)
 - Anatomical fit requirement
3. **Porous Coating**
 - Thickness: $0.030 \pm 0.010''$ ($0.76 \pm 0.25\text{mm}$)
 - Porosity: 40-60%
 - Bonding integrity
4. **Biocompatibility Surface**
 - No contamination
 - Specific surface energy
 - Cleanroom processing

Inspection Strategy:

Taper Measurement: - CMM with precision sphere probe - Helical scan pattern (200 points) - Least-squares fit to ideal taper - Calculate effective diameter at gauge planes - Measure angle deviation - Acceptance: All points within tolerance zone

Surface Finish: - Contact profilometer (cleanroom compatible) - Multiple circumferential locations - Average Ra must meet specification - Individual readings within 25% of average

Porous Coating: - Optical microscopy for thickness (cross-section sample) - CT scanning for porosity analysis - Pull-off testing for bond strength (destructive)

Dimensional Inspection: - CMM measurement of overall geometry - Profile comparison to CAD model - Documentation includes measurement uncertainty

Cleanliness Verification: - Wipe test and microscopic inspection - Total organic carbon (TOC) analysis - Particle count per ISO 14644

Documentation Requirements (per 21 CFR Part 820):

- Device Master Record (DMR)
- Device History Record (DHR) for each implant
- Inspection and test records with serial numbers
- Calibration records for all instruments
- Operator qualification records
- Material certifications and traceability
- Statistical process control data

Traceability: - Unique Device Identifier (UDI) for each implant - Complete genealogy from raw material to patient - Lifetime record retention (30+ years typical)

Outcome: - FDA approval maintained - Zero recalls due to dimensional defects - Full traceability for post-market surveillance - Process capability $C_{pk} > 2.0$ for critical features

21.12.3 Automotive Manufacturing Quality Control

High-volume automotive manufacturing requires fast, automated inspection integrated into production lines.

Case Study: Engine Block Machining Cell

Component: Aluminum engine block **Production:** 400 blocks per day **Cycle Time:** 8 minutes per block

Critical Features:

1. **Cylinder Bores** (4 cylinders)
 - Diameter: $86.000 +0.030/-0.000$ mm
 - Perpendicularity to deck: 0.02mm
 - Position tolerance: 0.05mm at MMC
 - Circularity: 0.01mm
 - Surface finish: Ra 0.4 μ m
2. **Main Bearing Bores** (5 bores)
 - Diameter: 52.000 ± 0.010 mm
 - Coaxiality: 0.02mm
 - Bore-to-bore spacing: ± 0.02 mm
3. **Deck Surface**
 - Flatness: 0.05mm across surface
 - Surface finish: Ra 1.6 μ m
 - Datum feature A
4. **Coolant Passages**
 - Must be clear and connected
 - No internal casting defects

In-Line Inspection System:

Automated Measurement Station:

Air Gaging (Cylinder Bores): - Pneumatic probes inserted in each bore - Measures diameter continuously as probe traverses - Detects taper, out-of-round, and size - Measurement time: 15 seconds for all 4 bores - Resolution: 0.001mm - Data logged to SPC system automatically

Coordinate Measuring Machine (Bearings and Datum): - Dedicated inspection CMM adjacent to line - Every 10th block automatically routed to CMM - Full GD&T inspection (20 minutes) - Automated program (no operator required) - Results compared to tolerance automatically - Out-of-spec blocks routed to quarantine

Vision System (Deck Surface): - High-resolution cameras inspect deck - Automated defect detection (scratches, gouges) - Flatness measured with laser profilometry - 100% inspection (5 seconds per block)

Pressure Test (Coolant Passages): - Automated leak test - Pressure decay method - Detects cracks and porosity - 100% tested - 30 seconds per block

SPC Implementation:

Real-Time Monitoring: - X-bar and R charts for bore diameter - Automatic control limits - Alarm if out of control - Trending analysis

Process Control Actions:

1. **In Control:** Continue production
2. **Trending toward limit:** Adjust tool offsets preemptively
3. **Out of control:** Stop line, investigate root cause
4. **Single out-of-spec part:** Quarantine, continue monitoring

Tool Management: - Automatic tool wear compensation - Tool change triggered by SPC data or preset limits - Tool life optimization (not premature replacement)

Results: - **Scrap rate:** < 0.1% (vs. 2-3% before SPC) - **Rework rate:** < 0.5% - **Process Cpk:** 1.67 average (exceeds 1.33 target) - **Machine utilization:** > 85% - **Quality cost reduction:** 40% over 2 years

Key Success Factors: - Fast automated inspection (no bottleneck) - Real-time SPC with automatic response - Integration with machine controls - Operator training and empowerment - Continuous improvement culture

21.12.4 High-Precision Tooling Measurement

Precision tooling requires ultra-tight tolerances and meticulous measurement.

Case Study: Injection Mold Cavity

Application: High-volume optical lens mold **Features:** Complex curved surfaces, mirror finish **Tolerances:** +/-0.005mm form, Ra 0.01 mum (10nm)

Challenges: - Sub-micron form accuracy required - Optical surface finish (mirror polish) - No witness marks or contact damage acceptable - Reverse-engineered from master lens

Measurement Approach:

White Light Interferometry: - Non-contact measurement (critical for polished surface) - Sub-nanometer resolution - 3D surface map of cavity - Stitching multiple fields of view for complete surface - Measurement time: 30 minutes (including setup and stitching)

Process: 1. Clean cavity with optical-grade solvents 2. Mount mold on stable platform (vibration-isolated table) 3. Align interferometer to cavity surface 4. Capture multiple overlapping measurements 5. Stitch data to create complete surface 6. Best-fit to nominal CAD surface 7. Generate deviation map

Results Analysis: - Form deviation: Peak +/-0.003mm (within +/-0.005mm spec) - Surface finish: Ra 8nm (within 10nm spec) - No scratches or defects detected - Data archived for process validation

CMM Verification (Non-Optical Features): - Mounting faces and alignment features - Cavity position relative to parting line - Cooling channel positions - Overall mold dimensions

In-Process Monitoring: - Mold inspection after every 50,000 shots - Track wear and degradation - Plan maintenance before failure - Maintain production quality

Outcome: - Mold produces optical-quality lenses - First article meets all specifications - Production maintained >500,000 shots before refurbishment - Customer approval for high-value application

21.12.5 Troubleshooting Measurement Discrepancies

Measurement discrepancies between different methods or operators are common challenges requiring systematic investigation.

Case Study: CMM vs. Micrometer Disagreement

Problem Statement: Machinist measures shaft diameter as 1.2505" with micrometer (within tolerance of 1.250 +/-0.002"). QC inspector measures same shaft on CMM as 1.2525" (out of upper tolerance). Part rejected. Machinist disputes. Who is correct?

Investigation Process:

Step 1: Verify Calibration - Check micrometer calibration certificate: Due date valid, last calibration passed - Check CMM calibration: Annual calibration current, probe qualification valid - Conclusion: Both instruments calibrated

Step 2: Examine Measurement Technique

Micrometer: - Observed operator technique: Measuring at one location, moderate force - Issue identified: Measuring at only one axial location

CMM: - Reviewed CMM program: Circle fit from 8 points at one Z height - Issue identified: Only one cross-section measured

Step 3: Comprehensive Measurement

CMM - Multiple Cross-Sections: - Measured circles at 5 axial locations along shaft - Results: - Z = 0mm: Ø1.2523" - Z = 10mm: Ø1.2518" - Z = 20mm: Ø1.2515" (center) - Z = 30mm: Ø1.2508" - Z = 40mm: Ø1.2505"

Conclusion: Shaft has taper of 0.0018" over length (hourglass shape)

Step 4: Root Cause Analysis

Machining Process Review: - Part held in 3-jaw chuck - Long slender shaft ($L/D = 10$) - Single pass turning operation

Root Cause: Workpiece deflection during cutting - Cutting forces pushed part away from tool at center - Result: Smaller diameter at ends, larger at center - Taper: 0.0018" over length

Step 5: Corrective Action

Immediate: - Parts measured at location of maximum diameter - Out-of-tolerance parts identified and scrapped or reworked

Permanent Fix: - Added center support (steady rest) during turning - Reduced cutting forces (lighter DOC, optimized speed/feed) - Changed cutting sequence (rough from both ends, finish pass)

Process Validation: - CMM measurement of revised process - Cylindricity improved from 0.0018" to 0.0004" - All cross-sections within tolerance - Cpk improved from 0.6 to 1.8

Lessons Learned:

1. **Multiple measurement locations:** Single-point measurements can miss form errors
2. **Understand GD&T:** Drawing didn't specify cylindricity tolerance (assumed straightness)
3. **Communication:** Machinist and inspector measuring different things (different locations)
4. **Root cause:** Both instruments were correct; part had form error
5. **Process improvement:** Metrology data revealed machining problem

Updated Inspection Procedure: - CMM measures cylindricity (multiple cross-sections) - Inspection drawing clarified with cylindricity tolerance - Operator training on proper measurement technique - Added cylindricity monitoring to SPC

Module Summary

Module 21 has covered the comprehensive field of precision measurement and metrology for CNC machining. From fundamental concepts of accuracy and traceability to advanced technologies like CT scanning and interferometry, metrology provides the foundation for quality manufacturing.

Key Principles: - Measurement is integrated throughout the manufacturing process - Proper technique and environmental control are as important as instrument quality - GD&T provides unambiguous communication of requirements - Statistical methods enable process control and improvement - Documentation and traceability are essential for quality systems

Practical Applications: - Every industry has unique metrology requirements - Automated in-line inspection enables high-volume manufacturing - Advanced technologies solve previously impossible measurement challenges - Systematic troubleshooting resolves measurement discrepancies

Path Forward: - Master basic measurement tools and techniques first - Understand GD&T thoroughly (essential for modern manufacturing) - Learn SPC for process monitoring and improvement - Develop systematic inspection planning skills - Stay current with evolving metrology technologies

Final Review Questions

1. In the turbine blade case study, why was CT scanning necessary in addition to CMM inspection?
 2. What regulatory requirements drive medical device inspection documentation?
 3. How did the automotive engine block case achieve such low scrap rates?
 4. Why was white light interferometry chosen for the optical mold cavity measurement?
 5. In the shaft taper case study, what was the root cause of the measurement discrepancy?
 6. What is the difference between measuring at one cross-section versus full cylindricity?
 7. How does SPC enable proactive quality control versus reactive inspection?
 8. Why is traceability critical in aerospace and medical device manufacturing?
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Practical Exercises

Exercise 1: Create an inspection plan for a simple part drawing with GD&T callouts. Include: - Feature list with specifications - Measurement method for each feature - Instrument requirements - Datum establishment procedure - Sample size and frequency

Exercise 2: Calculate measurement uncertainty for a dimensional measurement: - Calibration uncertainty: $\pm 0.003\text{mm}$ - Repeatability (from Gage R&R): $\pm 0.005\text{mm}$ - Temperature effect: $\pm 0.002\text{mm}$ - Resolution: 0.001mm

Exercise 3: Analyze an X-bar and R chart dataset to determine if the process is in statistical control. Identify any out-of-control conditions using Western Electric Rules.

Exercise 4: Calculate process capability (C_p and C_{pk}) from measurement data and determine if the process meets typical automotive requirements ($C_{pk} \geq 1.33$).

Congratulations on completing Module 21 - Metrology and Precision Measurement!

Precision measurement is the foundation of quality manufacturing. The skills and knowledge developed in this module will serve you throughout your career in CNC engineering and beyond.

Module 21 - Metrology and Precision Measurement

21.2.1 International Standards (ISO, ASME, DIN)

Measurement standards provide a common language for dimensional verification across industries and nations. Three major standards systems dominate precision manufacturing:

ISO (International Organization for Standardization)

The ISO system is used globally, particularly in Europe and Asia.

Key ISO Standards for Metrology:

- **ISO 1101:** Geometrical Product Specifications (GPS) - defines GD&T symbols and concepts
- **ISO 286:** System of limits and fits - standardizes hole and shaft tolerances
- **ISO 1:** Standard reference temperature of 20°C for length measurements
- **ISO 3650:** Gage blocks specifications
- **ISO 4287:** Surface texture parameters (Ra, Rz, etc.)
- **ISO 10360:** Coordinate Measuring Machine (CMM) acceptance tests
- **ISO 14253:** Decision rules for proving conformance with specifications

ASME (American Society of Mechanical Engineers)

ASME standards are dominant in North America.

Key ASME Standards:

- **ASME Y14.5:** Dimensioning and Tolerancing - the definitive North American GD&T standard
- **ASME B89.1.9:** Gage blocks
- **ASME B89.4.1:** Coordinate measuring machines
- **ASME B46.1:** Surface texture parameters
- **ASME Y14.43:** Dimensioning and tolerancing principles for gages and fixtures

DIN (Deutsches Institut für Normung)

German standards that influenced both ISO and ASME systems.

Key DIN Standards:

- **DIN 862:** Calipers
- **DIN 863:** Micrometers
- **DIN 2257-1:** Gage blocks
- **DIN 4768:** Surface roughness

21.2.2 Gage Blocks and Reference Standards

Gage blocks are the fundamental physical length standards used in shops and calibration labs.

What Are Gage Blocks?

Gage blocks are precision ground rectangular blocks of hardened steel, carbide, or ceramic with: - Two extremely flat, parallel measuring faces - Dimensions accurate to millionths of an inch - Surfaces so flat they can “wring” together (adhere via molecular attraction)

Gage Block Grades

ASME B89.1.9 / ISO 3650 Grades:

Grade	Tolerance	Application
Grade 0.5	+/(0.5 + L/400) mum	Reference standards in calibration labs
Grade 1	+/(0.8 + L/400) mum	Laboratory masters
Grade 2	+/(1.0 + L/400) mum	Inspection room use
Grade 3	+/(1.5 + L/400) mum	Shop floor setups

21.2.3 Calibration Hierarchies and Traceability

Calibration is the process of comparing a measurement instrument against a known reference standard and documenting the results.

The Calibration Hierarchy

Level 1 - Primary Standards: Maintained by national metrology institutes (NIST, NPL, PTB)

Level 2 - Secondary Standards: Maintained by accredited calibration laboratories

Level 3 - Working Standards: Used in quality labs and inspection rooms

Level 4 - Production Instruments: Used for daily production measurement

21.2.4 Calibration Intervals and Documentation

Calibration intervals balance cost with risk. Common intervals: - Precision instruments (CMM probes, laser systems): 6-12 months - Shop micrometers and calipers: 12 months - Working gage blocks: 1-3 years - Reference standards: 1-5 years

21.2.5 Temperature Compensation (20°C Reference)

ISO 1 establishes **20°C (68°F)** as the standard reference temperature for all dimensional measurements.

Linear Thermal Expansion Formula:

$$\Delta L = \alpha \times L \times \Delta T$$

Where: - ΔL = Change in length - α = Coefficient of thermal expansion (CTE) - L = Original length
- ΔT = Change in temperature from reference

Coefficients of Thermal Expansion

Material	CTE (ppm/°C)
Aluminum	23.0
Steel (carbon)	11.5
Stainless steel	17.3
Titanium	8.6
Granite	8.0

Module 21 - Metrology and Precision Measurement

21.3.1 Micrometers and Calipers

Micrometers

The micrometer is one of the most fundamental precision measurement tools in machining.

Types of Micrometers: - **Outside Micrometer:** Measures external dimensions - **Inside Micrometer:** Measures internal dimensions (bores, slots) - **Depth Micrometer:** Measures depth of holes, slots, recesses - **Thread Micrometer:** Measures thread pitch diameter - **Blade Micrometer:** Thin anvils for narrow slots and grooves

Resolution: - Standard: 0.001" (0.01mm) - Vernier: 0.0001" (0.002mm) - Digital: 0.00005" (0.001mm)

Reading a Micrometer:

1. **Sleeve:** Marked in 0.025" increments (imperial) or 0.5mm (metric)
2. **Thimble:** 25 divisions (imperial) or 50 divisions (metric)
3. **Vernier** (if equipped): Adds additional decimal place

Proper Micrometer Technique:

- Use ratchet stop or friction thimble for consistent pressure
- Measure at multiple points and average
- Clean measuring faces before use
- Allow thermal stabilization
- Check zero reading before use
- Store in protective case

Common Errors: - **Parallax error:** Reading scale at an angle - **Excessive force:** Deforming part or instrument - **Dirt on anvils:** False high readings - **Temperature difference:** Between part and standard

Calipers

Calipers are versatile instruments for measuring external, internal, and depth dimensions.

Types: - **Vernier Caliper:** Mechanical, 0.001" resolution - **Dial Caliper:** Mechanical with dial indicator, 0.001" resolution - **Digital Caliper:** Electronic display, 0.0005" resolution

Measurement Capabilities: - Outside jaws: External dimensions - Inside jaws: Internal dimensions (bores, slots) - Depth rod: Depth measurements - Step measurement: Using both beams

Proper Caliper Technique:

- Close jaws with light, consistent pressure
- Ensure jaws are perpendicular to surface
- For inside measurements, rock caliper to find maximum dimension
- Clean jaws and surfaces before measuring
- Avoid using excessive force
- Check zero before use

Limitations: - Less accurate than micrometers - Prone to misalignment - Easier to damage - Battery drain in digital models

Best Practices: - Use micrometers for critical dimensions - Use calipers for quick checks and non-critical features - Verify critical measurements with multiple instruments

21.3.2 Dial and Digital Indicators

Indicators measure relative displacement with high resolution and are essential for setup and inspection.

Dial Indicators

Components: - Plunger with spherical or flat contact point - Rack and pinion mechanism - Dial face with graduations - Mounting stem or back

Specifications: - **Travel:** Typically 0.5" to 2" (10mm to 50mm) - **Resolution:** 0.001" or 0.0001" (0.01mm or 0.002mm) - **Accuracy:** +/-0.001" over full range

Common Applications: - Runout measurement (TIR - Total Indicator Reading) - Part alignment and setup - Machine tool calibration - Surface flatness checking - Comparative measurement

Dial Test Indicator (DTI):

Also called "finger indicator" or "interapid" (brand name).

- Lever-actuated contact point
- Contact point can be positioned at various angles
- Better for confined spaces
- More sensitive to off-axis loading

Resolution: 0.0005" or 0.0001" (0.01mm or 0.002mm)

Digital Indicators

Modern electronic indicators with digital displays.

Advantages: - Easier to read (no parallax error) - Data output capability (SPC integration) - Zeroing at any position - Inch/metric switchable - Memory functions

Disadvantages: - Battery dependent - More expensive - Can be affected by electrical interference

Indicator Mounting

Magnetic Base: - Strong permanent magnet - On/off switch - Articulating arms - Fine adjustment knob

Other Mounting Methods: - Clamp mounts - Dovetail attachments - Universal holders - Post mounts on surface plates

Proper Indicator Technique

Setup: 1. Mount indicator rigidly 2. Position contact perpendicular to surface 3. Preload plunger to mid-range 4. Zero or note starting position

Measurement: - Rotate part slowly for runout checks - Take readings at multiple positions - Record maximum and minimum values - $TIR = \text{Maximum} - \text{Minimum reading}$

Common Errors: - **Cosine error:** Contact not perpendicular - **Excessive plunger extension:** Beyond linear range - **Loose mounting:** Affects repeatability - **Dirty contact:** False readings

21.3.3 Height Gages and Depth Gages

Height Gages

Height gages measure vertical distances from a reference surface (typically a surface plate).

Types: - **Vernier Height Gage:** Mechanical, 0.001" resolution - **Dial Height Gage:** With dial indicator - **Digital Height Gage:** Electronic, 0.0005" resolution - **Electronic Height Gage:** With motorized carriage

Components: - Base (sits on surface plate) - Vertical beam with scale - Scriber or indicator attachment - Fine adjustment wheel

Applications: - Layout and scribing operations - Comparative height measurement - Center-to-center distance measurement - Flatness and parallelism checks (with indicator)

Proper Technique:

1. Use precision ground surface plate as reference
2. Clean base and plate surfaces
3. Ensure base sits flat (rock gently to check)
4. Zero at plate surface or use known standard
5. Move scribe/indicator perpendicular to measured surface
6. For multiple measurements, maintain consistent technique

Depth Gages

Measure depth of holes, slots, recesses, and step heights.

Types: - Depth micrometer - Vernier depth gage - Digital depth gage - Dial depth gage

Technique: - Use flat reference surface or gage block stack - Ensure base is perpendicular to hole axis - Measure at multiple points in hole - Watch for burrs at hole entrance - Support base on three points minimum

21.3.4 Bore Gages and Inside Micrometers

Bore Gages (Dial Bore Gage)

Highly accurate instruments for measuring hole diameters.

Components: - Measuring head with spring-loaded anvils - Interchangeable anvils for different size ranges - Dial or digital indicator - Master setting rings or gage blocks for zeroing

Size Ranges: - Small bore: 0.250" - 2.0" (6-50mm) - Standard: 2.0" - 6.0" (50-150mm) - Large bore: 6.0" - 12.0"+ (150-300mm+)

Resolution: 0.0001" or 0.0005" (0.002mm or 0.01mm)

Proper Technique:

1. **Setup:** Zero gage using master ring or gage block stack
2. **Insertion:** Insert gage into bore at slight angle
3. **Centering:** Rock gage side-to-side and front-to-back
4. **Reading:** Minimum reading indicates true diameter
5. **Multiple Points:** Measure at multiple depths and orientations

Common Measurements: - Bore diameter - Bore taper (measure at multiple depths) - Bore out-of-round (measure at 90° increments) - Bore bell-mouth (compare entry vs. deep readings)

Inside Micrometers

Extend micrometer capability to internal measurements.

Types: - **Three-point internal micrometer:** Self-centering - **Two-point internal micrometer:** Requires rocking technique - **Rod-type inside micrometer:** Interchangeable extension rods

Technique: - Rock micrometer to find maximum reading (true diameter) - Ensure measuring points contact on diameter (not chord) - Use light touch - Verify with known standard

21.3.5 Thread Gages and Pitch Micrometers

Thread Plug and Ring Gages

Go/No-Go Gages: - **GO gage:** Should thread smoothly (max material condition) - **NO-GO gage:** Should not thread (min material condition) - If GO threads and NO-GO doesn't, thread is within tolerance

Types: - Plug gage: Tests internal threads - Ring gage: Tests external threads

Limitations: - Only indicates pass/fail, not actual dimensions - Separate gages needed for each thread specification - Subject to wear

Thread Pitch Micrometers

Measures thread pitch diameter directly.

Features: - Pointed anvil and spindle shaped to fit thread form - Different anvil sets for different pitch ranges - Resolution: 0.001" (0.01mm)

Technique: - Select correct anvil for thread pitch - Align anvil with thread axis - Thread should rest in V-shaped anvil - Take multiple readings around circumference

Thread Wires (Three-Wire Method)

Precision method for measuring pitch diameter using calibrated wires.

Process: 1. Select wire size appropriate for thread pitch 2. Place two wires in thread on one side, one on opposite side 3. Measure over wires with micrometer 4. Calculate pitch diameter using formula

Advantages: Very accurate, works with any thread form

Disadvantages: Time-consuming, calculation required, difficult setup

21.3.6 Surface Plates and Granite Fixtures

Precision Surface Plates

Surface plates provide a flat reference datum for measurement and layout.

Materials: - **Granite:** Most common, excellent stability, non-magnetic - **Cast Iron:** Traditional, can be refinished, magnetic (useful for fixtures) - **Ceramic:** Lightweight, excellent thermal stability

Grades (Federal Specification GGG-P-463c):

Grade	Flatness Tolerance (24" x 24" plate)
AA Laboratory	0.000050"
A Inspection	0.000100"
B Tool Room	0.000200"

Sizes: From 12" x 18" to 72" x 144" and larger

Surface Plate Care

Do's: - Cover when not in use - Clean regularly with appropriate solvent - Use soft brush or lint-free cloth - Place work gently - Use parallels and blocks to distribute load - Calibrate annually (grade dependent)

Don'ts: - Never strike parts on plate - Don't drag heavy parts across surface - Avoid using near grinding operations (abrasive dust) - Don't store tools or parts on plate - Never use as assembly bench

Granite Squares and Fixtures

Precision Squares: - 90° reference standards - Grades: AA, A, B - Used for setup and inspection

Angle Plates: - Precision right-angle fixtures - Mounting holes for work holding - Used with surface plate for perpendicular measurement

V-Blocks: - Hold cylindrical parts for measurement - Matched pairs for shaft measurement - Typically 90° V-angle

Summary

Basic measurement tools form the foundation of shop floor metrology. Mastering these instruments—micrometers, calipers, indicators, height gages, bore gages, and surface plates—is essential for any precision machinist. Understanding proper technique, sources of error, and appropriate applications ensures accurate measurements and quality parts.

Key Takeaways

1. **Micrometers** provide highest accuracy for linear dimensions (0.0001")
 2. **Calipers** are versatile but less accurate than micrometers
 3. **Indicators** measure relative displacement and are essential for setup
 4. **Height gages** require precision surface plates as reference
 5. **Bore gages** give most accurate internal diameter measurements
 6. **Thread gages** provide quick pass/fail inspection
 7. **Surface plates** must be maintained and calibrated regularly
 8. **Proper technique** is as important as instrument quality
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Review Questions

1. What is the difference between a dial indicator and a dial test indicator?
2. Why should you use the ratchet stop on a micrometer?
3. How do you properly measure a bore diameter with a dial bore gage?
4. What does TIR stand for and how is it calculated?
5. Explain the GO/NO-GO gage concept for thread inspection.
6. Why are granite surface plates preferred over cast iron in most modern shops?
7. Calculate: If a dial indicator reads +0.008" at the maximum point and -0.003" at the minimum point during a runout check, what is the TIR?

Module 21 - Metrology and Precision Measurement

21.4.1 Coordinate Measuring Machines (CMM)

The Coordinate Measuring Machine (CMM) is the cornerstone of modern precision inspection, capable of measuring complex 3D geometries with micron-level accuracy.

21.4.1.1 CMM Types and Configurations

Bridge CMM (Most Common): - Fixed table with moving bridge - Three linear axes (X, Y, Z) - High accuracy and rigidity - Sizes from 500mm to several meters - Typical accuracy: $1.5\text{-}3.0\text{ }\mu\text{m} + L/300$

Gantry CMM: - Open structure with moving table - Suitable for large, heavy parts - Used in automotive and aerospace - Ground-level loading

Cantilever CMM: - L-shaped structure - Easy access for loading - Less rigid than bridge type - Good for small to medium parts

Horizontal Arm CMM: - Horizontal probe orientation - Ideal for large automotive body panels - Easy part access

Portable/Articulated Arm CMM: - Handheld articulated arm - Brought to part instead of part to CMM - Lower accuracy but highly flexible - Useful for large assemblies or field measurement

Vision CMM: - Optical/video-based measurement - Non-contact, ideal for soft parts - Limited to 2D or surface features

CMM Accuracy Specifications

Volumetric Accuracy:

Typically expressed as: $E = A + L/K$

Where: - E = Maximum permissible error - A = Constant error (instrument accuracy) - L = Measured length - K = Constant (typically 200-400)

Example: $1.5\text{ }\mu\text{m} + L/300$ means: - 100mm measurement: $\pm 1.8\text{ }\mu\text{m}$ - 500mm measurement: $\pm 3.2\text{ }\mu\text{m}$

Repeatability: Typically 1/3 to 1/2 of accuracy specification

21.4.1.2 Probe Systems (Touch, Scanning, Optical)

Touch Trigger Probes:

- Most common CMM probe type
- Makes point contact with part
- Sends signal when probe deflects threshold amount
- Single points are measured and processed

Advantages: - Reliable and robust - Relatively inexpensive - Easy to program - Good for discrete features (holes, edges, spheres)

Disadvantages: - Slow for complex surfaces - Point density limited - Contact force may deform soft parts

Scanning Probes (Analog/Continuous):

- Maintains continuous contact while moving
- Captures thousands of points per second
- Creates dense point clouds

Types: - **Analog scanning:** Measures probe deflection continuously - **Passive scanning:** Part moves through stationary probe - **Active scanning:** Probe actively follows surface

Advantages: - Very fast data acquisition - Excellent for freeform surfaces - High point density

Disadvantages: - More expensive - Requires sophisticated software - More complex programming

Optical/Laser Probes:

- Non-contact measurement
- Uses laser triangulation or structured light
- Ideal for soft, delicate, or complex parts

Advantages: - No contact force (won't deform part) - Extremely fast data capture - Can measure complex organic shapes

Disadvantages: - Surface finish affects accuracy - Reflective or transparent surfaces problematic - Less accurate than touch probes

Probe Changing Systems:

- Automatic probe changing rack
- Allows multiple probe configurations in one program
- Different stylus lengths, angles, and types
- Greatly increases measurement flexibility

21.4.1.3 CMM Programming and Operation

Manual CMM Operation:

- Operator guides probe using joystick
- Software prompts for feature type
- Useful for one-off inspections
- Teaches basics of CMM measurement

DCC (Direct Computer Control):

- CMM follows programmed path automatically
- Most efficient for production inspection
- Consistent and repeatable
- Requires programming time investment

Programming Methods:

1. **Teach Mode:** Manually guide probe, record positions
2. **CAD Programming:** Import CAD model, select features to measure
3. **Offline Programming:** Program without using CMM (based on CAD)

Measurement Strategy:

Datum Establishment: 1. Measure primary datum feature first (typically a plane) 2. Measure secondary datum (perpendicular plane or axis) 3. Measure tertiary datum (complete constraint)

Feature Measurement: - Planes: Minimum 3 points (more for large or irregular surfaces) - Circles: Minimum 4 points (8-12 typical for accuracy) - Cylinders: Multiple circles at different heights - Spheres: Minimum 4 points (more for better accuracy)

Probe Qualification (Probe Calibration):

Before measurement, probe tip must be calibrated: 1. Measure reference sphere at multiple orientations 2. Software calculates effective probe tip diameter 3. Compensates for probe deflection and stylus geometry 4. Required after probe change or crash

Environmental Control:

- CMMs require stable temperature (20°C +/- 1°C)
- Vibration isolation (isolated foundation)
- Clean environment (dust affects accuracy)
- Part soaking time: 1 hour per inch of thickness

Best Practices:

- Clean part before measurement (oil, chips, dirt affect results)
 - Use minimum probing force for delicate parts
 - Measure critical features multiple times
 - Document measurement uncertainty
 - Regular CMM calibration using certified standards
-

21.4.2 Laser Interferometry

Laser interferometers are the most accurate method for measuring linear displacement and are used to calibrate machine tools and CMMs.

Principle of Operation

Interference of Light Waves:

- Laser beam split into reference and measurement beams
- Measurement beam reflects off moving target
- Beams recombine creating interference fringes
- Each fringe represents $\lambda/2$ displacement (typically ~316nm for HeNe laser)
- Electronic counting of fringes gives displacement

Accuracy: 0.1 ppm (parts per million) of reading

Example: Over 1 meter, accuracy is +/-0.1 μm

Applications

Machine Tool Calibration: - Measures linear positioning accuracy - Identifies systematic errors - Creates error compensation maps

CMM Calibration: - Verifies volumetric accuracy - Meets ISO 10360 requirements

Stage and Encoder Calibration: - Reference standard for rotary encoders - Linear stage verification

Length Measurement: - Precision length standards - Gage block calibration

Environmental Corrections

Laser wavelength depends on air conditions:

Corrections Applied: - Temperature - Pressure - Humidity - CO₂ content

Modern systems measure environmental parameters and automatically compensate.

21.4.3 Linear and Rotary Encoders

Encoders provide position feedback in CNC machines and measurement systems.

Linear Encoders

Types:

Incremental Encoders: - Output pulses for each increment of movement - Require homing to establish position - Resolution: 0.1 μm to 5 μm typical - Most common in CNC machines

Absolute Encoders: - Provide unique position code at every point - No homing required - More expensive - Critical in systems that can't lose position

Technologies:

Optical Encoders: - Glass scale with etched graduations - LED and photodetector read position - High accuracy ($\pm 0.5 \mu\text{m}$ to $\pm 5 \mu\text{m}$) - Sensitive to contamination

Magnetic Encoders: - Magnetic tape scale - Hall effect or magnetoresistive sensor - More robust in harsh environments - Lower accuracy than optical

Rotary Encoders

Measure angular position of rotating shafts.

Applications: - Spindle position (for rigid tapping, thread milling) - Rotary table position - Tool changer indexing

Resolution: Expressed in pulses per revolution (PPR) - Low resolution: 100-1000 PPR - High resolution: 10,000-100,000 PPR

21.4.4 Optical Comparators and Vision Systems

Optical Comparators

Project magnified shadow of part onto screen for visual inspection.

Components: - Light source (transmitted or reflected) - Precision lens system - Projection screen with overlay or digital readout - XY measuring stage

Magnifications: 5x, 10x, 20x, 50x, 100x

Applications: - Profile inspection (gears, threads, stamped parts) - Screw thread inspection - Small part inspection - Comparing to master overlay

Advantages: - Visual comparison intuitive - Fast for complex profiles - Non-contact measurement

Disadvantages: - Limited to 2D profiles - Subjective (operator dependent) - Edge detection affected by lighting and finish

Vision Measurement Systems

Modern computerized vision systems with automated edge detection and measurement.

Technologies: - High-resolution digital cameras - LED illumination (transmitted, reflected, coaxial) - Motorized XY stage - Automated focus (Z-axis)

Measurement Software: - Automatic edge detection - Geometric element fitting (circles, lines, arcs) - GD&T analysis - CAD comparison - Statistical reporting

Advantages: - Fast automated measurement - Objective and repeatable - 3D capability (with multiple cameras or laser) - Data output for SPC - CAD overlay and comparison

Applications: - Electronics (PCB inspection, connector pins) - Medical devices (small components) - Plastic parts (soft, easily deformed) - Stamped parts (complex 2D profiles) - Multi-cavity mold quality

21.4.5 Surface Finish Measurement (Profilometry)

Surface finish significantly affects part function, wear, friction, and appearance.

Contact Profilometers

Stylus Profilometer: - Diamond stylus (radius 2-10 μm typical) - Traverses surface at constant speed - Vertical displacement recorded - Analog or digital output

Parameters Measured: - R_a (average roughness) - R_z (average peak-to-valley height) - R_q (RMS roughness) - Profile length and sampling

Standards: ISO 4287, ASME B46.1

Advantages: - Well-established technology - Traceable to standards - Works on most materials

Disadvantages: - Contact can scratch soft surfaces - Slow measurement - 2D profile only (not true 3D surface)

Non-Contact Profilometry

Optical Profilometers:

Technologies: - **White light interferometry:** Sub-nanometer resolution, ideal for very smooth surfaces - **Confocal microscopy:** Good for rough and textured surfaces - **Laser triangulation:** Fast, suitable for rougher surfaces

Advantages: - Non-contact (won't damage surface) - 3D surface maps - Very fast measurement - Can measure features invisible to stylus (steep walls, re-entrant features)

Disadvantages: - Expensive - Surface properties affect measurement (reflectivity, transparency) - Calibration more complex

Surface Finish Parameters

Amplitude Parameters: - **R_a :** Average roughness (most common parameter) - **R_z :** Average of highest peaks and lowest valleys - **R_q :** RMS roughness - **R_t :** Total height (maximum peak to valley)

Spacing Parameters: - **R_{Sm} :** Mean spacing of profile elements - **Peak count:** Number of peaks per unit length

Best Practices: - Measure multiple locations on surface - Orient measurement perpendicular to lay (tool marks) - Specify evaluation length appropriate to feature - Clean surface before measurement - Record measurement conditions

Summary

Advanced measurement instruments—CMMs, laser interferometers, encoders, optical comparators, and profilometers—provide capabilities far beyond basic hand tools. CMMs measure complex 3D geometry with micron accuracy. Laser systems calibrate machines to sub-micron precision. Vision systems automate inspection. Profilometers quantify surface texture. Mastering these instruments is essential for modern precision manufacturing.

Key Takeaways

1. **CMMs** are the primary tool for complex 3D inspection
 2. **Touch probes** are reliable; **scanning probes** are faster for complex surfaces
 3. **Laser interferometry** provides the highest accuracy for linear measurement
 4. **Encoders** provide real-time position feedback in CNC systems
 5. **Vision systems** automate 2D measurement and inspection
 6. **Profilometers** quantify surface finish objectively
 7. **Environmental control** is critical for all advanced instruments
 8. **Probe qualification** must be performed before CMM measurement
-

Review Questions

1. What is the difference between a touch trigger probe and a scanning probe on a CMM?
2. Calculate the maximum permissible error for a CMM with specification $2.0 \text{ } \mu\text{m} + L/300$ when measuring a 400mm dimension.
3. Why must a CMM probe be “qualified” before measurement?
4. What is the principle of operation for a laser interferometer?
5. What is the difference between incremental and absolute encoders?
6. When would you choose an optical comparator over a CMM?
7. What does the Ra surface finish parameter represent?
8. Why are optical profilometers preferred over stylus instruments for very smooth surfaces?

Module 21 - Metrology and Precision Measurement

On-machine measurement integrates inspection directly into the machining process, reducing setup time, improving accuracy, and enabling adaptive machining strategies.

21.5.1 Touch Probes and Tool Setters

Touch Probes (Workpiece Probing)

Touch probes mounted in the machine spindle measure part location, features, and dimensions without removing the part.

Types:

Kinematic (Strain Gage) Probes: - Most accurate type - Measure deflection in X, Y, Z directions - Accuracy: ± 1 -2 μm - Typically optical or radio signal transmission - Examples: Renishaw RMP, Blum TC

Touch Trigger Probes: - Binary contact detection (touch/no touch) - Simpler and less expensive - Accuracy: ± 2 -5 μm - Examples: Renishaw OMP, Heidenhain TT

Signal Transmission: - **Infrared optical:** Line of sight required, no interference - **Radio:** 360° coverage, may have interference - **Hard-wired:** Most reliable but limits motion

Probe Applications

Part Setup and Alignment:

1. **Workpiece Location:** Find edges and corners to establish work coordinate system
2. **Rotation Alignment:** Detect and correct angular misalignment
3. **Feature Location:** Find hole centers, boss locations, slot positions

In-Process Inspection:

1. **Dimensional Verification:** Measure critical features before removing part
2. **Tool Wear Compensation:** Detect excessive tool wear and adjust
3. **Multi-Setup Verification:** Verify datum alignment between operations

Adaptive Machining:

1. **Stock Measurement:** Measure actual stock condition (casting variation)
2. **Dynamic Offset:** Adjust tool offsets based on measured conditions
3. **Closed-Loop Machining:** Re-measure and re-machine until tolerance achieved

Probe Calibration

Probe Qualification:

Before use, probe must be calibrated to establish: - Effective tip diameter - Trigger point in all directions - Stylus length

Process: 1. Mount reference sphere (certified master) 2. Probe sphere from multiple angles (typically 5-25 points) 3. Control calculates effective probe diameter and offsets 4. Store calibration data in machine

Re-qualification Required: - After probe/stylus change - After probe crash or suspected damage - Periodically (daily for critical work) - After machine warmup

Tool Setters

Tool setters measure tool length and diameter automatically.

Types:

Spindle-Mounted Setter: - Fixed to machine column or table - Tools lower to contact sensor - Measures tool length offset (Z)

Rotating Tool Setter: - Laser or optical sensor - Measures tool diameter as tool rotates - Detects broken or chipped tools

Applications:

1. **Tool Length Measurement:** Automatic tool offset setting
2. **Tool Breakage Detection:** Verify tool before and after machining
3. **Tool Wear Monitoring:** Track dimensional wear over time
4. **Pre-Setting Verification:** Confirm pre-set tool dimensions

Benefits: - Eliminates manual touch-off errors - Reduces setup time significantly - Enables lights-out manufacturing - Provides tool condition monitoring

21.5.2 In-Process Measurement Systems

In-process measurement occurs during actual machining, providing real-time feedback.

Gauging Probes

In-Cycle Gauging:

- Probe integrated into machining cycle
- Measures while part is still being machined
- Provides immediate feedback for offset adjustment

Post-Process Gauging:

- Measures after machining operation completes
- Verifies dimensions before part release
- Can trigger automatic rework cycle if out of tolerance

Integrated Measurement Systems

Laser Micrometers:

- Non-contact diameter measurement
- Measures rotating parts (grinding, turning)
- Accuracy: $\pm 1-3 \text{ }\mu\text{m}$
- Real-time feedback for grinding operations

Air Gauging:

- Pneumatic measurement system
- Measures bore diameters in-process
- Very fast response time
- Excellent for high-volume production

LVDT (Linear Variable Differential Transformer):

- Analog position sensor
- Continuous monitoring during machining
- Used in precision grinding
- Micron-level resolution

Benefits of In-Process Measurement

1. **Reduced Scrap:** Catch errors before completion
 2. **Faster Cycles:** No separate inspection operation
 3. **Tool Wear Compensation:** Automatically adjust for wear
 4. **Process Monitoring:** Track trends, predict failures
 5. **Statistical Data:** Automatic SPC data collection
-

21.5.3 Adaptive Machining with Closed-Loop Feedback

Adaptive machining uses measurement feedback to automatically adjust machining parameters.

Closed-Loop Machining Process

Traditional (Open-Loop): 1. Program assumes nominal stock and tool conditions 2. Machine executes program 3. Inspect after completion 4. Scrap or rework if out of tolerance

Adaptive (Closed-Loop): 1. Measure actual stock condition 2. Calculate required material removal 3. Machine with adjusted parameters 4. Measure result 5. Repeat until within tolerance

Applications

Casting Machining:

Problem: Castings have significant dimensional variation

Solution: - Probe casting surfaces before machining - Map actual stock location - Adjust toolpaths to accommodate variation - Optimize material removal

High-Value Part Finishing:

Problem: Cannot scrap expensive forgings or castings

Solution: - Machine conservatively - Measure in-process - Take additional finishing passes if needed - Guarantee first-piece success

Thin-Wall Machining:

Problem: Deflection causes dimensional errors

Solution: - Measure after roughing passes - Adjust semi-finishing passes based on results - Final light finishing pass after stress relief

Implementation Considerations

Advantages: - Reduced scrap on first article - Accommodates stock variation - Enables near-net-shape manufacturing - Improves process capability

Disadvantages: - Longer cycle time (measurement time) - Complex programming - Requires skilled setup and maintenance - Higher equipment cost

21.5.4 Tool Wear Monitoring

Tool wear directly affects part quality and dimensional accuracy. Monitoring tool condition prevents scrapped parts.

Direct Tool Measurement

Tool Setter Monitoring:

- Measure tool before and after machining
- Compare to known new tool dimensions
- Set wear limits (warning and reject)
- Automatic tool change when limit reached

Typical Monitoring: - Tool length: Detects breakage or excessive wear - Tool diameter: Detects chipping or gradual wear - Runout: Detects holder problems

Indirect Tool Monitoring

Power/Load Monitoring:

- Monitor spindle load or servo current
- Dull tools require more cutting force
- Set thresholds for tool change
- Can detect tool breakage instantly

Vibration Monitoring:

- Accelerometers detect vibration signatures
- Chipped or broken tools create abnormal vibration
- Can differentiate tool problems from other sources

Acoustic Emission:

- Ultrasonic sensors detect high-frequency signals
- Tool fracture creates distinctive signature
- Very fast response (milliseconds)

Tool Life Management

Tool Life Counters:

- Track cutting time or parts machined
- Replace tool at predetermined interval
- Conservative but simple

Adaptive Tool Life:

- Adjust tool life based on actual wear measurement
- Maximize tool utilization
- Prevent premature replacement

Tool Management Software:

- Database of tool inventory
 - Track tool life history
 - Predict tool replacement needs
 - Optimize tool usage across multiple machines
-

21.5.5 Workpiece Setup and Datum Probing

Accurate part alignment is fundamental to precision machining. Probing automates and improves setup accuracy.

Workpiece Location (Part Finding)

Edge Finding:

- Probe part edges in X and Y
- Establish work coordinate system origin
- Accuracy: $\pm 0.002\text{--}0.005\text{ mm}$ ($\pm 0.0001\text{--}0.0002''$)

Process: 1. Probe X+ edge (approach from X-) 2. Probe X- edge (approach from X+) 3. Calculate X center: $(X+ + X-) / 2$ 4. Repeat for Y axis 5. Set work coordinate system (G54-G59)

Corner Finding:

- Probe two perpendicular edges
- Establish X, Y origin at corner
- Faster than full edge finding

Web/Boss Measurement:

- Measure internal web or external boss
- Establish center position
- Use 3-5 probe points for best accuracy

Angle/Rotation Measurement

Two-Point Rotation:

- Measure two points along part edge
- Calculate rotation angle
- Apply coordinate system rotation
- Tolerance: $\pm 0.01^\circ$ typical

Process: 1. Probe part edge at two Y positions 2. Calculate: $\theta = \arctan((X_2 - X_1) / (Y_2 - Y_1))$ 3. Set work coordinate system rotation

3D Datum Establishment:

Part Top Surface (Z Zero): - Probe top surface at multiple points - Average for tilt compensation - Set Z work offset

Datum Sequence (per GD&T): 1. Probe primary datum (plane or cylinder) 2. Probe secondary datum (orientation constraint) 3. Probe tertiary datum (location constraint) 4. Calculate work coordinate system transformation

Bore and Hole Center Finding

Circular Feature Probing:

- Probe hole or boss at 3-4 points minimum
- Calculate best-fit circle center
- Set X, Y work offset

Process (for hole): 1. Pre-position probe near hole center 2. Probe at 0° , 90° , 180° , 270° (or more points) 3. Calculate center: Fit circle to measured points 4. Set work coordinate system

Accuracy Considerations: - More probe points = better accuracy - Use odd number spacing for best results - 3 points minimum, 4-5 typical, 8+ for critical features

Multi-Face/Multi-Setup Probing

Tombstone Setup:

- Probe each part on multi-face fixture

- Establish individual work offsets (G54.1 P1-P99)
- Compensate for fixture tolerance stack-up

Between-Operation Datum:

- Probe finished features from prior operation
 - Establish datum for current operation
 - Ensures feature-to-feature alignment
-

Summary

On-machine measurement transforms CNC machining from an open-loop to closed-loop process. Touch probes automate setup and enable in-process inspection. Tool setters ensure dimensional accuracy. Adaptive machining accommodates variation. Tool wear monitoring prevents scrapped parts. Together, these technologies improve quality, reduce cycle time, and enable lights-out manufacturing.

Key Takeaways

1. **Touch probes** eliminate manual part setup and enable in-process inspection
 2. **Tool setters** automate tool measurement and detect breakage
 3. **Probe qualification** is required before measurement
 4. **Adaptive machining** uses closed-loop feedback to compensate for variation
 5. **Tool wear monitoring** prevents dimensional errors and scrap
 6. **Datum probing** establishes work coordinate systems automatically
 7. **In-process measurement** reduces cycle time and improves quality
 8. **On-machine measurement** enables lights-out manufacturing
-

Review Questions

1. What is the difference between a touch trigger probe and a strain gage probe?
2. Why must a touch probe be qualified before use?
3. Describe the process for finding the center of a hole using a touch probe.
4. What are three methods for indirect tool wear monitoring?
5. How does adaptive machining differ from traditional open-loop machining?
6. What is the advantage of using a tool setter versus manual tool touch-off?
7. List three applications for workpiece probing in CNC machining.
8. How does closed-loop machining reduce scrap for high-value castings?

Module 21 - Metrology and Precision Measurement

Geometric Dimensioning and Tolerancing (GD&T) is a symbolic language used on engineering drawings to define the allowable variation in part geometry. It provides a precise, unambiguous method for specifying design intent and inspection criteria.

21.6.1 GD&T Fundamentals

Why GD&T?

Traditional Plus/Minus Tolerancing Limitations:

- Ambiguous: Multiple interpretations possible
- Restrictive: Square tolerance zones waste manufacturing capability
- Incomplete: Doesn't address form, orientation, or location of features
- No datum precedence: Doesn't specify which surfaces establish reference frame

GD&T Advantages:

- **Precise:** Unambiguous communication of design requirements
- **Functional:** Tolerances reflect actual part function
- **Economical:** Larger tolerance zones where possible
- **International:** Standardized symbols understood globally
- **Inspection-friendly:** Clear measurement criteria

Basic GD&T Concepts

Feature: A specific portion of a part (surface, hole, slot, etc.)

Feature of Size: A feature with opposed points (cylinder, sphere, parallel planes)

Tolerance Zone: The boundary within which a feature must lie

Datum: A theoretically exact reference (plane, axis, or point) from which measurements are made

Datum Feature: The actual feature on the part used to establish the datum

Material Condition: The size of a feature of size (Maximum, Least, or Regardless)

GD&T Symbol Structure

Feature Control Frame:

A rectangular box divided into compartments containing:

Symbol	Tolerance	Datum A	Datum B	Datum C
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Example:

Position	Ø 0.010	A	B	C
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This means: Position tolerance of Ø0.010" at MMC, relative to datums A, B at MMC, and C at MMC.

21.6.2 Datums and Datum Reference Frames

Datums establish the coordinate system for measurement. Proper datum selection is critical for functional parts.

Datum Feature Selection

Primary Datum (First in sequence): - Constrains 3 degrees of freedom (typically a plane) - Usually the largest, most stable mating surface - Establishes orientation

Secondary Datum (Second in sequence): - Constrains 2 additional degrees of freedom - Perpendicular to primary (or at specified angle) - Establishes additional orientation constraint

Tertiary Datum (Third in sequence): - Constrains final degree of freedom - Completes 6-degree constraint (3 translations, 3 rotations) - Prevents rotation about secondary datum axis

Datum Feature Simulators

Datum simulators are the physical reference surfaces/axes used during inspection:

Plane Datum: Surface plate or precision angle plate

Cylindrical Datum: Precision pin or expanding mandrel

Width Datum: Parallel blocks or vise jaws

Datum Precedence

Datum order matters! A-B-C is different from A-C-B.

Example: Mounting bracket - **Datum A:** Mounting surface (plane) - how it attaches - **Datum B:** Dowel pin hole (axis) - prevents rotation - **Datum C:** Edge surface (plane) - prevents sliding

Datum Targets

For irregular or non-planar surfaces, datum targets specify exact contact points.

Datum Target Symbols: - Point: Single contact point - Line: Contact along a line - Area: Contact over defined area (circle or rectangle)

Application: Cast or forged parts with rough surfaces

21.6.3 Form Tolerances

Form tolerances control the shape of individual features without reference to datums.

Flatness

Symbol:

Definition: All points on the surface must lie between two parallel planes separated by the tolerance value.

Tolerance Zone: Two parallel planes

Example: 0.005 means the surface must lie within 0.005" wide zone

Measurement: - Place part on surface plate - Use height gage with indicator - Measure high and low points
- Flatness error = maximum variation

Refinement of: Refines the size tolerance of a width feature (doesn't require datum)

Straightness

Symbol: –

Definition: A line element on a surface must lie between two parallel lines separated by the tolerance value.

Two Applications:

Surface Straightness: -- 0.003 - Applies to surface elements - Tolerance zone: Two parallel lines - Checked using straightedge and feeler gage

Axis Straightness: $\emptyset$ 0.005 (cylindrical tolerance zone) - Applies to axis of cylindrical feature - Tolerance zone: Cylinder of diameter 0.005" - Allows feature to bow within tolerance

Measurement: Indicator sweep along surface or between centers for axis

Circularity (Roundness)

Symbol:

Definition: All points on a circular feature at any cross-section must lie between two concentric circles separated by the tolerance value (radial).

Tolerance Zone: Two concentric circles

Example: 0.002 means the radius variation cannot exceed 0.002" at any cross-section

Measurement: - Rotate part on precision spindle - Indicator records radial variation - Maximum variation = circularity error - Or measure on roundness tester

Not: Circularity does NOT control cylindricity (cross-sections can be parallel but offset)

Cylindricity

Symbol:

Definition: All points on a cylindrical surface must lie between two coaxial cylinders separated by the tolerance value (radial).

Tolerance Zone: Two coaxial cylinders

Example: 0.003 means the entire surface must lie within 0.003" radial zone

Controls: - Straightness of elements - Circularity of cross-sections - Taper - Out-of-round at any section

Measurement: CMM or cylindricity tester measuring helical or grid pattern

Strictest control: Most restrictive of all form controls

21.6.4 Orientation Tolerances

Orientation tolerances control angularity of features relative to datums.

Perpendicularity

Symbol:

Definition: A feature must be 90° to a datum within the specified tolerance.

Tolerance Zone: Two parallel planes perpendicular to datum

Example: 0.005 A means surface must be perpendicular to datum A within 0.005" wide zone

Applications: - Surface perpendicular to datum plane - Axis perpendicular to datum plane - Axis perpendicular to datum axis

Measurement: - Establish datum on surface plate - Indicator sweep perpendicular to datum - Maximum variation = perpendicularity error

Parallelism

Symbol:

Definition: A feature must be parallel to a datum within the specified tolerance.

Tolerance Zone: Two parallel planes parallel to datum

Example: 0.003 A means surface must be parallel to datum A within 0.003" wide zone

Applications: - Mating surfaces - Bearing mounts - Precision ways

Measurement: - Establish datum on surface plate - Height gage with indicator - Measure high and low points - Variation = parallelism error

Angularity

Symbol:

Definition: A feature must be at a specified angle to a datum within the tolerance.

Tolerance Zone: Two parallel planes at specified angle to datum

Example: 0.008 A at 45° means surface must be within 0.008" zone at 45° to datum A

Measurement: - Establish datum - Use precision angle block or sine plate - Indicator measures variation

21.6.5 Location Tolerances

Location tolerances control the position of features relative to datums or other features.

Position

Symbol:

Definition: A feature's location must be within a tolerance zone centered at the theoretically exact (true) position.

Tolerance Zone: - Cylindrical (for holes, pins): $\varnothing 0.010$ A B C - Width (for slots, tabs): 0.010 A B C (two parallel planes)

True Position: The theoretically exact location defined by basic dimensions (boxed dimensions on drawing)

Formula (for holes):

$$\text{Position error} = 2 \times \sqrt{(X_{\text{actual}} - X_{\text{true}})^2 + (Y_{\text{actual}} - Y_{\text{true}})^2}$$

Example: Hole at true position X0.000, Y0.000 measured at X0.003, Y0.004

$$\text{Error} = 2 \times \sqrt{(0.003)^2 + (0.004)^2} = 2 \times 0.005 = 0.010"$$

Bonus Tolerance: When position tolerance specifies MMC (), additional tolerance is gained as feature departs from MMC

Benefits: - Circular tolerance zone vs. square (57% more area) - Functional: Reflects mating conditions - Bonus tolerance increases yield

Concentricity

Symbol:

Definition: The median points of diametrically opposed elements of a feature must lie within a cylindrical tolerance zone coaxial with datum axis.

Tolerance Zone: Cylinder centered on datum axis

Example: 0.005 A means median points must lie within Ø0.005" cylinder coaxial with datum A

Application: Dynamic balance, rotating assemblies

Measurement: Difficult - requires measuring median points, not surface

Modern Practice: Often replaced by runout or position at RFS

Symmetry

Symbol:

Definition: The median points of opposed elements of a feature must lie within a tolerance zone centered on datum plane.

Tolerance Zone: Two parallel planes symmetrically disposed about datum plane

Example: 0.008 A means median points within 0.008" zone centered on datum plane A

Application: Symmetric parts, balanced features

Modern Practice: Rarely used; often replaced by position

21.6.6 Profile Tolerances

Profile tolerances control the form, orientation, and sometimes location of complex surfaces.

Profile of a Surface

Symbol:

Definition: The surface must lie between two boundaries offset equally (unless otherwise specified) from the true profile.

Tolerance Zone: 3D zone normal to true profile surface

Applications: - Airfoils and aerodynamic surfaces - Sculptured surfaces - Complex contours - Can control form, orientation, and location simultaneously

With Datum Reference: 0.010 A B C - Controls form, orientation, AND location

Without Datum Reference: 0.010 - Controls form only

Unequally Disposed: Can specify offset (e.g., +0.008/-0.002)

Measurement: CMM with CAD comparison

Profile of a Line

Symbol:

Definition: Each line element on the surface must lie between two boundaries in the plane of the line element.

Difference from Surface Profile: Controls line elements independently (no control between lines)

Application: Extrusions, constant cross-sections, less restrictive than surface profile

21.6.7 Runout

Runout controls the combined effects of form and location relative to a datum axis.

Circular Runout

Symbol:

Definition: Each circular element must be within tolerance when the part is rotated 360° about a datum axis.

Tolerance Zone: Two concentric circles at each measuring position

Example: 0.003 A means each cross-section must indicate ≤ 0.003 when rotated about datum A

Measurement: - Mount part on datum axis (between centers, on mandrel, or in collet) - Rotate part 360°
- Indicator at fixed position records variation - Repeat at multiple axial positions

Controls (at each position independently): - Circularity - Coaxiality - Perpendicularity (for face runout)

Total Runout

Symbol:

Definition: Entire surface must be within tolerance when part is rotated 360° about datum axis while indicator traverses the surface.

Tolerance Zone: Two coaxial cylinders (or two parallel planes for face)

Example: 0.005 A means entire surface must indicate ≤ 0.005 during full rotation and traverse

Measurement: - Mount part on datum axis - Rotate part continuously - Traverse indicator along surface - Maximum variation = total runout error

Controls: - Circular runout at all positions - Cylindricity - Straightness - Taper - Coaxiality

Most Comprehensive: Total runout is the most restrictive control for rotating parts

21.6.8 Material Condition Modifiers

Material condition modifiers adjust tolerance based on feature size.

Maximum Material Condition (MMC) -

Definition: The condition where a feature contains the maximum amount of material within size limits.

Examples: - Shaft: Largest allowable diameter - Hole: Smallest allowable diameter

Bonus Tolerance Concept:

When MMC is specified, additional tolerance is allowed as the feature departs from MMC.

Example: Hole $\varnothing 0.500 \pm 0.005$, Position $\varnothing 0.010$ A B C

Actual Hole Size	Position Tolerance	Total Tolerance
0.495 (MMC)	0.010	0.010
0.500 (Nominal)	0.015	0.015

Actual Hole Size	Position Tolerance	Total Tolerance
0.505 (LMC)	0.020	0.020

Bonus = Actual size - MMC size

Functional Justification: Larger hole accepts larger positional error while maintaining assembly

Benefits: - Increased manufacturing tolerance - Reduced scrap - Reflects functional requirements

Least Material Condition (LMC) -

Definition: The condition where a feature contains the minimum amount of material within size limits.

Examples: - Shaft: Smallest allowable diameter - Hole: Largest allowable diameter

Application: Controls minimum wall thickness, minimum material in threaded features

Less Common: Used primarily when minimum material is critical

Regardless of Feature Size (RFS)

Definition: Geometric tolerance applies at any feature size; no bonus tolerance.

Default in ASME Y14.5-2009: If no modifier shown, RFS is implied

Application: When function requires consistent geometric tolerance regardless of size

Summary

GD&T provides a precise, functional, and internationally recognized language for specifying and verifying part geometry. Form tolerances control shape. Orientation tolerances control angles relative to datums. Location tolerances control position. Profile controls complex shapes. Runout controls rotating features. Material condition modifiers optimize tolerance. Mastering GD&T is essential for modern precision manufacturing.

Key Takeaways

1. **GD&T** eliminates ambiguity in engineering drawings
 2. **Datums** establish the reference frame for measurement
 3. **Datum precedence** (A-B-C order) matters
 4. **Form tolerances** (flatness, straightness, circularity, cylindricity) don't require datums
 5. **Orientation tolerances** (perpendicularity, parallelism, angularity) require datums
 6. **Position** is the most common and versatile location control
 7. **Profile** controls complex surfaces in 3D
 8. **Total runout** is the most restrictive control for rotating parts
 9. **MMC modifier** provides bonus tolerance and increases manufacturing yield
 10. **GD&T enables** larger tolerances while maintaining function
-

Review Questions

1. What is the difference between a datum and a datum feature?
2. Why does datum precedence matter?

3. Which form tolerance is most restrictive for a cylindrical shaft?
4. Calculate the position error for a hole at true position X0.000, Y0.000, measured at X-0.004, Y+0.003.
5. What is the difference between circular runout and total runout?
6. A $\varnothing 0.250 +0.005/-0.000$ hole has position tolerance $\varnothing 0.008$. What is the total position tolerance when the hole measures $\varnothing 0.253$?
7. When would you use profile of a line versus profile of a surface?
8. What does the MMC modifier do and why is it beneficial?

Module 21 - Metrology and Precision Measurement

Statistical Process Control (SPC) uses statistical methods to monitor and control manufacturing processes, enabling early detection of problems and continuous improvement.

21.7.1 Control Charts and Process Capability

What is SPC?

SPC is a methodology for: - Monitoring process variation over time - Distinguishing normal (common cause) variation from abnormal (special cause) variation - Detecting process shifts before defects occur - Providing objective data for process improvement

Control Charts

Control charts plot measurement data over time and compare it to statistical control limits.

Basic Components:

1. **Center Line (CL)**: Process average
2. **Upper Control Limit (UCL)**: $CL + 3\sigma$
3. **Lower Control Limit (LCL)**: $CL - 3\sigma$
4. **Data Points**: Measurements plotted chronologically

3-Sigma Limits: ± 3 standard deviations from mean - Captures 99.73% of normal variation - Points outside limits indicate “out of control” process

Types of Control Charts

Variable Data Charts (Continuous measurements):

X-bar and R Chart (Average and Range): - Most common SPC chart - X-bar chart: Plots sample averages - R chart: Plots sample ranges - Detects shifts in process mean and variation

X-bar and S Chart (Average and Standard Deviation): - Similar to X-R chart - S chart plots standard deviation - Better for sample sizes > 10

Individual and Moving Range (I-MR) Chart: - For individual measurements (sample size = 1) - Used when subgrouping is impractical - Moving range tracks consecutive differences

Attribute Data Charts (Count/Pass-Fail data):

p Chart: Proportion defective (variable sample size)

np Chart: Number defective (constant sample size)

c Chart: Count of defects per unit (constant sample size)

u Chart: Defects per unit (variable sample size)

Control Limits Calculation

X-bar and R Chart Example:

Given: 25 subgroups of $n=5$ measurements

Step 1: Calculate average of each subgroup ($\bar{X}$)

Step 2: Calculate range of each subgroup ($R = \max - \min$)

Step 3: Calculate grand average and average range:

$\bar{\bar{X}} = \sum \bar{X} / \text{number of subgroups}$

$\bar{R} = \sum R / \text{number of subgroups}$

Step 4: Calculate control limits using constants:

X-bar Chart:

$$UCL = \bar{\bar{X}} + A_2 \times \bar{R}$$

$$CL = \bar{\bar{X}}$$

$$LCL = \bar{\bar{X}} - A_2 \times \bar{R}$$

R Chart:

$$UCL = D_4 \times \bar{R}$$

$$CL = \bar{R}$$

$$LCL = D_3 \times \bar{R}$$

Constants (for n=5): - $A_2 = 0.577$ - $D_3 = 0$ - $D_4 = 2.114$

Interpreting Control Charts

In Control: Process exhibits only random variation - All points within control limits - Random pattern around center line - No trends or patterns

Out of Control: Special cause variation present

Rules for Detecting Special Causes (Western Electric Rules):

1. **One point** beyond $\pm 3\sigma$ control limits
2. **Two of three consecutive points** beyond $\pm 2\sigma$ (on same side)
3. **Four of five consecutive points** beyond $\pm 1\sigma$ (on same side)
4. **Eight consecutive points** on one side of center line
5. **Six consecutive points** trending up or down
6. **Fourteen consecutive points** alternating up and down
7. **Fifteen consecutive points** within $\pm 1\sigma$ (hugging center)

Actions When Out of Control: - Stop production and investigate - Identify root cause of special variation
- Implement corrective action - Resume production and continue monitoring

21.7.2 Cp, Cpk, and Process Performance Indices

Process capability indices quantify how well a process meets specifications.

Process Capability Index (Cp)

Cp measures the potential capability of a process (if perfectly centered).

Formula:

$$Cp = (USL - LSL) / (6\sigma)$$

Where: - USL = Upper Specification Limit - LSL = Lower Specification Limit - σ = Process standard deviation - 6σ = Natural process spread (99.73% of distribution)

Interpretation:

Cp Value	Interpretation	Capability
$Cp < 1.0$	Process spread > spec range	Not capable
$Cp = 1.0$	Process spread = spec range	Minimally capable
$Cp = 1.33$	Process spread = 75% of spec	Adequate
$Cp = 1.67$	Process spread = 60% of spec	Good
$Cp \geq 2.0$	Process spread = 50% of spec	Excellent

Limitation: Cp doesn't account for process centering

Process Capability Index (Cpk)

Cpk measures actual capability considering process centering.

Formula:

$$Cpk = \min[(USL - \mu)/(3\sigma), (\mu - LSL)/(3\sigma)]$$

Where: μ = Process mean - σ = Process standard deviation

Alternative Formula:

$$Cpk = Cp \times (1 - k)$$

where $k = |\mu - \text{Target}| / (0.5 \times \text{Tolerance})$

Interpretation:

Cpk Value	Expected Defect Rate	Interpretation
Cpk = 0.67	4.5%	Poor
Cpk = 1.0	0.27% (2700 ppm)	Marginal
Cpk = 1.33	63 ppm	Acceptable (industry standard)
Cpk = 1.67	0.6 ppm	Good
Cpk = 2.0	0.002 ppm	Excellent (Six Sigma)

Cpk vs. Cp: - If $Cpk = Cp$: Process is perfectly centered - If $Cpk < Cp$: Process is off-center - Goal: Maximize Cpk by centering process

Process Performance Indices (Pp, Ppk)

Pp and **Ppk** are similar to Cp and Cpk but use overall standard deviation (not within-subgroup).

Difference: - **Cp/Cpk:** Use within-subgroup variation (short-term capability) - **Pp/Ppk:** Use overall variation (long-term performance)

Comparison: - If $Pp \approx Cp$: Process is stable over time - If $Pp \ll Cp$: Process drifts or has special causes

Calculating Standard Deviation for Capability

Short-Term (Cp, Cpk): Use within-subgroup variation

$$\sigma = R\text{-bar} / d_2$$

Where d_2 is constant based on sample size ($d_2 = 2.326$ for $n=5$)

Long-Term (Pp, Ppk): Use overall sample standard deviation

$$\sigma = \sqrt{[\sum(x - \bar{x})^2 / (n-1)]}$$

Example Calculation

Given: - Specification: 25.000 $\pm$ 0.050 mm (USL=25.050, LSL=24.950) - Process mean: $\mu = 25.010$ mm
- Process std dev: $\sigma = 0.015$ mm

Calculate Cp:

$$Cp = (25.050 - 24.950) / (6 \times 0.015)$$

$$Cp = 0.100 / 0.090 = 1.11$$

Calculate Cpk:

$$CPU = (25.050 - 25.010) / (3 \times 0.015) = 0.040 / 0.045 = 0.89$$

$$CPL = (25.010 - 24.950) / (3 \times 0.015) = 0.060 / 0.045 = 1.33$$

$$Cpk = \min(0.89, 1.33) = 0.89$$

Interpretation: - Cp = 1.11: Process spread nearly fills spec (marginal if centered) - Cpk = 0.89: Process is off-center toward upper limit (not capable) - Action: Center process to improve Cpk

21.7.3 Sampling Plans and Inspection Strategies

Sampling Plans

100% Inspection vs. Sample Inspection:

100% Inspection: - Every part measured - Used for critical features, small lots, high-value parts - Expensive and time-consuming - Risk of inspection errors (fatigue)

Sample Inspection: - Subset of parts measured - Lower cost - Risk of accepting bad lots or rejecting good lots - Appropriate for high-volume production

Acceptance Sampling

Single Sampling Plan: - Take sample of n parts - Measure/inspect - Accept lot if defects $\leq$ acceptance number (c) - Reject lot if defects $> c$

Example: n=50, c=2 - Inspect 50 parts - Accept if ≤ 2 defects found - Reject if ≥ 3 defects found

Double Sampling Plan: - Take first sample (n₁) - Accept if defects $\leq c_1$ - Reject if defects $\geq c_2$
- Take second sample (n₂) if between c₁ and c₂ - Make final decision based on combined sample

Sampling Standards

ANSI/ASQ Z1.4 (formerly MIL-STD-105E): - Statistical sampling tables - Based on lot size and acceptable quality level (AQL) - Provides normal, tightened, and reduced inspection plans

ANSI/ASQ Z1.9: - Similar to Z1.4 but for variable data (measurements) - Generally smaller sample sizes than attribute plans

Risk in Sampling

Producer's Risk (alpha): Probability of rejecting a good lot - Type I error - Typically set at 5%

Consumer's Risk (beta): Probability of accepting a bad lot - Type II error - Typically set at 10%

Operating Characteristic (OC) Curve: - Graphically shows probability of acceptance vs. lot quality - Used to evaluate sampling plan performance

Inspection Strategies

Incoming Inspection: - Verify supplier quality - Critical for purchased components - Can be reduced with supplier certification

In-Process Inspection: - Catch problems early - SPC at critical operations - First article, last article, and periodic sampling

Final Inspection: - Verify finished product - Typically more comprehensive - Documentation for shipment

Skip-Lot Inspection: - Inspect only fraction of lots from proven supplier - Reduces cost while maintaining quality assurance

21.7.4 Measurement System Analysis (MSA)

Before trusting measurement data for SPC or process capability, the measurement system itself must be validated.

Components of Measurement Variation

Total Observed Variation = Part Variation + Measurement Variation

Measurement Variation: - **Repeatability:** Same operator, same part, same instrument (Equipment Variation - EV) - **Reproducibility:** Different operators, same part, same instrument (Appraiser Variation - AV) - **Linearity:** Accuracy across measurement range - **Bias:** Systematic offset from true value - **Stability:** Consistency over time

21.7.4.1 Gage R&R Studies

Gage Repeatability and Reproducibility (R&R) study quantifies measurement system variation.

Standard AIAG Method:

Setup: - 10 parts spanning the process range - 3 operators - Each operator measures each part 3 times (in random order) - Total: 90 measurements (10 x 3 x 3)

Calculations:

Repeatability (EV):

$$EV = (R/d_2) \times K_1$$

Where R = average range within parts

Reproducibility (AV):

$$AV = \sqrt{[(X_{diff} \times K_2)^2 - (EV^2/nr)]}$$

Where Xdiff = range of operator averages

Gage R&R:

$$GRR = \sqrt{(EV^2 + AV^2)}$$

%GRR (Relative to Tolerance):

$$\%GRR = (GRR / \text{Tolerance}) \times 100\%$$

Interpretation:

%GRR	Rating	Action
< 10%	Excellent	Acceptable
10-30%	Acceptable	May be acceptable depending on application
> 30%	Unacceptable	Improvement required

Alternative: %GRR relative to total variation

Actions for Poor Gage R&R: - High repeatability (EV): Calibrate equipment, improve maintenance, reduce environmental variation - High reproducibility (AV): Improve operator training, standardize technique, use fixtures - Both high: Consider different measurement method or instrument

21.7.4.2 Bias and Linearity Studies

Bias Study:

Tests if measurement system has systematic offset.

Process: 1. Measure reference standard multiple times 2. Calculate average measured value 3. Compare to known standard value 4. Bias = Average measured - True value

Acceptance: Bias should be < 5% of tolerance

Linearity Study:

Tests if bias is consistent across the measurement range.

Process: 1. Select 5-10 reference standards spanning range 2. Measure each standard multiple times 3. Calculate bias at each point 4. Plot bias vs. reference value 5. Calculate linearity (slope of bias line)

Acceptance: Linearity < 5% of tolerance

Action if Failed: Calibrate instrument, check for wear, consider different instrument

Summary

Statistical Process Control provides objective methods for monitoring and improving manufacturing processes. Control charts detect process shifts in real-time. Process capability indices (Cp, Cpk) quantify how well processes meet specifications. Sampling plans balance inspection cost with risk. Measurement System Analysis ensures measurement data is reliable. Together, these tools enable data-driven quality control and continuous improvement.

Key Takeaways

1. **Control charts** distinguish common cause from special cause variation
 2. **3-sigma limits** capture 99.73% of normal variation
 3. **X-bar and R charts** are most common for variable data
 4. **Cp** measures potential capability (centered process)
 5. **Cpk** measures actual capability (accounts for centering)
 6. **Cpk** ≥ 1.33 is typical industry minimum requirement
 7. **Gage R&R studies** validate measurement system capability
 8. **%GRR** < 10% is excellent; >30% requires action
 9. **Sampling plans** balance cost and risk
 10. **SPC is proactive:** Prevents defects rather than detecting them
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Review Questions

1. What is the difference between common cause and special cause variation?
2. Calculate the UCL and LCL for an X-bar chart where $\bar{X} = 25.00$, $\bar{R} = 0.12$, $n = 5$, $A_2 = 0.577$.
3. A process has $USL=50.0$, $LSL=40.0$, $\mu=45.5$, $\sigma=1.5$. Calculate Cp and Cpk.
4. What does it mean if Cp is much larger than Cpk?
5. What is the difference between Cp/Cpk and Pp/Ppk?
6. A Gage R&R study shows %GRR = 35%. What does this mean and what action is required?
7. List three rules that indicate a process is out of statistical control.
8. Why is Cpk = 1.33 commonly used as a minimum requirement?

Module 21 - Metrology and Precision Measurement

Machine tool accuracy directly affects part quality. Regular assessment and calibration ensure machines maintain specified performance over time.

21.8.1 ISO 230 Standard for Machine Tool Testing

ISO 230 is the international standard for testing the accuracy of machine tools.

ISO 230 Structure

ISO 230-1: General concepts, definitions, and test methods

ISO 230-2: Determination of accuracy and repeatability of positioning

ISO 230-3: Determination of thermal effects

ISO 230-4: Circular tests for CNC machine tools

ISO 230-6: Determination of positioning accuracy on body and face diagonals (ballbar)

ISO 230-7: Geometric accuracy of axes of rotation

Key Concepts

Accuracy: Closeness of commanded position to actual position

Repeatability: Variation when returning to same position multiple times

Resolution: Smallest increment system can distinguish

Reversal Error (Backlash): Position error when approaching from opposite directions

Positioning Deviation: Difference between target and actual position

Test Conditions

Environmental Requirements: - Stable temperature (typically 20°C +/-2°C) - Thermal equilibrium (machine warm-up) - No external vibration - Consistent humidity

Machine Preparation: - Run machine for minimum 30 minutes at typical duty cycle - Lubrication system operational - No abnormal wear or damage

21.8.2 Geometric Accuracy (21 Error Components)

Every machine axis has multiple error components that affect overall accuracy.

The 21 Geometric Errors (for 3-axis machine)

For Each Linear Axis (X, Y, Z) - 6 errors per axis:

1. **Linear positioning error:** EXX, EYY, EZZ
 - Deviation along its own axis
2. **Horizontal straightness:** EXY, EYZ, EZX
 - Deviation perpendicular to motion (horizontal)
3. **Vertical straightness:** EXZ, EYX, EYZ
 - Deviation perpendicular to motion (vertical)
4. **Roll:** EXA, EYB, EZC
 - Rotation about axis of motion
5. **Pitch:** EXB, EYC, EZA

- Rotation about horizontal perpendicular axis
6. **Yaw:** EXC, EYA, EZB
- Rotation about vertical perpendicular axis

Total: 18 errors (3 axes x 6 errors each)

Additional 3 Squareness Errors:

7. **Squareness between axes:**
- SXY: X to Y squareness
 - SYZ: Y to Z squareness
 - SZX: Z to X squareness

Total: 21 geometric errors

Measuring Geometric Errors

Linear Positioning: - Laser interferometer along axis - Multiple positions across full travel - Bi-directional measurement

Straightness: - Laser with straightness optics - Optical level or autocollimator - Across full axis travel

Angular Errors (Roll, Pitch, Yaw): - Electronic levels - Autocollimators - Laser with angular optics

Squareness: - Laser interferometer with multiple reflectors - Granite square and indicators - CMM verification

Error Mapping and Compensation

Modern CNC controls can apply **volumetric error compensation:**

1. Measure all 21 error components
2. Create error map across machine volume
3. Control applies real-time corrections
4. Significantly improves machine accuracy

Typical Improvement: 50-80% reduction in positioning errors

21.8.3 Ballbar Testing for Circular Interpolation

The ballbar test (per ISO 230-4) evaluates circular interpolation accuracy using a telescoping bar with precision ball joints.

Ballbar Setup

Components: - Telescoping bar with linear encoder or LVDT - Precision ball ends (kinematic coupling) - Center mount (magnetic base on machine table) - Spindle mount (ball cup attached to spindle or Z-axis)

Operation: 1. Mount center ball on table 2. Attach spindle ball to machine 3. Program circular interpolation (typically Ø300mm) 4. Machine executes circle while ballbar measures radius variation 5. Software analyzes results

What Ballbar Tests

Circular Interpolation: - Synchronization between X and Y axes - Servo mismatch - Backlash and reversal spikes - Stick-slip friction - Scale mismatch

Benefits: - Quick test (15-30 minutes per orientation) - Highly sensitive to machine problems - Visual graphical output - Trend tracking over time

Interpreting Ballbar Results

Ideal Result: Perfect circle (all deviations near zero)

Common Error Signatures:

Circular Form (Ovality): - Squareness error between axes - Scale mismatch between axes - Perpendicularity issue

Reversal Spikes: - Backlash in drives or ballscrews - Clearance in couplings - Lost motion in system

Quadrant Glitches: - Servo tuning issues - Friction variation between directions - Stick-slip behavior

Vibration/Lobing: - Mechanical resonance - Servo instability - Structural issues

Typical Acceptance: Circularity within ± 5 -10 μm for precision machines

Ballbar Test Planes

Horizontal Plane (XY): - Most common test - Z-axis at mid-travel

Vertical Planes (XZ, YZ): - Tests different axis combinations - Important for 5-axis machines

Multiple Radii and Feedrates: - Different radii reveal scale errors - Different feedrates test servo performance

21.8.4 Laser Calibration Systems

Laser interferometry provides the most accurate method for calibrating machine tool positioning.

Linear Axis Calibration

Setup: 1. Mount laser unit to machine table or bed 2. Attach retroreflector to spindle or Z-axis 3. Align laser beam parallel to axis 4. Input environmental compensation (temperature, pressure, humidity)

Test Process: 1. Program axis to move to specified positions 2. Laser measures actual position at each point 3. Bi-directional measurement (forward and reverse) 4. Multiple runs for statistical validation

Measurements Obtained: - **Positioning error:** Deviation from commanded position - **Repeatability:** Standard deviation at each position - **Reversal error:** Difference between forward and reverse approach - **Hysteresis:** Non-repeatable errors

Typical Grid: 20-30 positions across axis travel

Angular Calibration

Optical Square or Pentaprism: - Measures squareness between axes - Accuracy: ± 0.5 arcseconds

Angular Optics: - Laser with angular measurement capability - Measures roll, pitch, yaw - Resolution: 0.1 arcseconds

Rotary Axis Calibration

Laser or Electronic Autocollimator: - Measures angular positioning of A/B/C axes - Polygon or rotary indexer as reference

Typical Accuracy: ± 1 -2 arcseconds for precision rotary tables

Calibration Frequency

New Machine Acceptance: Full geometric calibration

Periodic Verification: - Annual: Production machines - Semi-annual: Precision machines - Quarterly: Ultra-precision machines - After crash or major repair

Quick Checks: Ballbar testing monthly or quarterly

21.8.5 Thermal Drift and Compensation

Temperature changes cause machine structures and components to expand, creating positioning errors.

Sources of Thermal Drift

Internal Heat Sources: - Spindle bearings and motor - Ballscrews and linear guides - Servo motors and drives - Hydraulic systems - Electrical cabinets

External Heat Sources: - Ambient temperature variation (day/night, HVAC) - Direct sunlight - Adjacent equipment - Personnel proximity

Coolant Temperature: Variations affect workpiece and tooling

Thermal Error Magnitude

Example: A 1-meter steel column with 1°C temperature rise expands 11.5 µm

Typical Thermal Drift: 10-50 µm over hours of operation

Impact: Can exceed machine's geometric accuracy by an order of magnitude!

Thermal Stabilization

Warm-Up Period: - Run machine at typical duty cycle - Typically 30-60 minutes minimum - Longer for high-precision work - Spindle warm-up critical

Environmental Control: - Temperature-controlled shop (+/-1-2°C) - Avoid direct sunlight on machines - Consistent HVAC operation - Thermal curtains or enclosures

Thermal Compensation

Passive Methods: - Thermally symmetric design - Low CTE materials (Invar, ceramics) - Thermal isolation of heat sources - Coolant through structures

Active Compensation:

Temperature Sensor Arrays: - Thermocouples or RTDs at multiple locations - Measure ballscrew, column, bed temperatures - Real-time monitoring

Compensation Models: - Mathematical model relates temperature to position error - Control applies real-time corrections - Typical improvement: 70-90% reduction in thermal errors

Adaptive Learning: - System learns thermal behavior over time - Improves compensation accuracy - Accounts for seasonal variations

ISO 230-3 Thermal Testing

Environmental Temperature Variation Test (ETVE): - Measures position error as ambient temperature varies - Machine at rest or low-duty cycle - Quantifies passive thermal stability

Thermal Distortion Test: - Run spindle and drives at high duty cycle - Monitor position drift over 4+ hours - Measures active thermal effects

21.8.6 Volumetric Error Mapping

Volumetric error is the combined effect of all 21 geometric errors at any point in the machine's work envelope.

Volumetric Accuracy Measurement

Traditional Method: - Measure all 21 error components individually - Computationally combine using kinematics - Time-consuming (days)

Laser Tracker or Laser Tracer: - Measures 3D position at grid points throughout volume - Fast (hours instead of days) - Directly measures volumetric error

Ballbar Body Diagonal Test (ISO 230-6): - Ballbar measurement along space diagonals - Tests three-axis coordinated motion - Reveals volumetric errors - Quick screening test

Volumetric Error Compensation

Process: 1. Measure volumetric errors at grid points 2. Create 3D error map 3. Fit mathematical model (polynomial or spline) 4. Load compensation table into control 5. Control applies corrections in real-time

Benefits: - Significantly improves accuracy across entire volume - Extends machine capability - Enables tighter tolerances - Can compensate aging machine

Limitations: - Cannot compensate dynamic errors (cutting forces) - Requires periodic recalibration - Complex setup and modeling

Summary

Machine tool accuracy assessment is fundamental to precision manufacturing. ISO 230 provides standardized test methods. The 21 geometric errors define machine accuracy. Ballbar testing quickly evaluates circular interpolation. Laser systems provide traceable calibration. Thermal effects often dominate errors and must be controlled or compensated. Volumetric error mapping and compensation significantly improve machine performance. Regular assessment ensures machines maintain specification over time.

Key Takeaways

1. **ISO 230** is the international standard for machine tool testing
 2. **21 geometric errors** define 3-axis machine accuracy
 3. **Repeatability** is typically 3-5x better than positioning accuracy
 4. **Ballbar testing** quickly evaluates circular interpolation and servo performance
 5. **Laser interferometry** provides highest accuracy for calibration
 6. **Thermal drift** can exceed geometric errors by 10x
 7. **Warm-up period** (30-60 min) is essential for accuracy
 8. **Volumetric compensation** improves accuracy 50-80%
 9. **Regular calibration** maintains machine performance over time
 10. **Thermal compensation** is critical for precision work
-

Review Questions

1. What is the difference between accuracy and repeatability in machine tool testing?
2. List the six error components for a single linear axis.
3. What does ballbar testing measure and what are common error signatures?
4. Why is thermal drift often the dominant error source in machine tools?
5. How long should a machine warm up before precision machining?
6. What is volumetric error and how is it different from individual axis errors?
7. Calculate the thermal expansion of a 500mm steel component with 2°C temperature rise ($\alpha = 11.5$ ppm/°C).
8. When should a production CNC machine undergo full geometric calibration?

Module 21 - Metrology and Precision Measurement

Surface finish affects part function, appearance, wear characteristics, fatigue life, corrosion resistance, and assembly fit. Measuring and controlling surface texture is essential for quality manufacturing.

21.9.1 Surface Roughness Parameters (Ra, Rz, Rq)

Surface texture consists of irregularities created by the manufacturing process. Multiple parameters quantify different aspects of the surface.

Surface Texture Components

Roughness: Fine irregularities from the manufacturing process (tool marks, feed marks) - Wavelength: < 0.8 mm typically - Amplitude: Micrometers to nanometers

Waviness: Longer wavelength deviations (machine vibration, workpiece deflection, heat treatment) - Wavelength: 0.8 mm to several mm - Amplitude: Typically larger than roughness

Form: Overall shape of the surface (flatness, straightness, roundness) - Wavelength: Entire part surface - Removed by filtering for roughness measurement

Lay: Predominant direction of surface pattern (parallel, perpendicular, circular, etc.)

Common Roughness Parameters

Ra (Arithmetic Average Roughness):

Most widely used parameter worldwide.

Definition: Average absolute deviation of roughness profile from mean line

Formula:

$$Ra = (1/L) \int_0^L |y(x)| dx$$

Where: - L = Evaluation length - y = Profile height deviation from mean line

Units: micrometers (µm), microinches (µin)

Typical Values: - Rough machining: Ra = 6-25 µm (250-1000 µin) - General machining: Ra = 1.6-6 µm (63-250 µin) - Fine machining: Ra = 0.4-1.6 µm (16-63 µin) - Grinding: Ra = 0.1-0.8 µm (4-32 µin) - Lapping: Ra = 0.01-0.1 µm (0.4-4 µin)

Advantages: Simple, well-understood, widely specified

Limitations: Doesn't distinguish peaks from valleys, insensitive to occasional high peaks

Rz (Average Maximum Height):

Definition (ISO 4287): Average of absolute values of five highest peaks and five lowest valleys within evaluation length

Formula:

$$Rz = (\sum |R_{pi}| + \sum |R_{vi}|) / 5$$

Where: - R_{pi} = Peak heights - R_{vi} = Valley depths

Alternative (older definition): Maximum peak-to-valley height (R_t or R_{max})

Advantages: More sensitive to extreme peaks/valleys than Ra

Applications: Important when peaks might scratch mating surfaces or when valleys affect sealing

Typical Relationship: Rz approx 4-8 x Ra (varies with process)

Rq (Root Mean Square Roughness):

Definition: RMS average of profile height deviations

Formula:

$$R_q = \sqrt{\left(1/L\right) \int_0^L y^2(x) dx}$$

Characteristics: - Always slightly larger than Ra (typically Rq approx 1.1 x Ra) - More sensitive to peaks and valleys than Ra - Used in theoretical analysis and optical measurements

Rt (Total Height):

Definition: Maximum peak-to-valley height over evaluation length

Characteristics: - Most severe parameter - Varies significantly with evaluation length - Sensitive to measurement noise and anomalies

Additional Parameters

RSm (Mean Spacing of Profile Irregularities): - Average spacing between peaks - Important for lubrication and wear

Rsk (Skewness): - Symmetry of profile - Positive: Sharp peaks - Negative: Deep valleys

Rku (Kurtosis): - Sharpness of profile - High: Spiky surface - Low: Bumpy surface

Evaluation Length and Cutoff

Cutoff Length (lambdac): Wavelength that separates roughness from waviness - Standard values: 0.08, 0.25, 0.8, 2.5, 8.0 mm - Selection based on expected roughness

Evaluation Length (ln): Total length over which parameters are calculated - Typically 5x cutoff length - $ln = 5 \times \text{lambdac}$

Example: For Ra = 1.6 µm surface - Cutoff: lambdac = 0.8 mm - Evaluation length: ln = 4.0 mm

21.9.2 Waviness and Form Separation

Proper filtering separates roughness, waviness, and form for meaningful measurement.

Filtering Process

Raw Profile: Contains form, waviness, and roughness

High-Pass Filter: Removes waviness and form, leaves roughness - Cutoff wavelength lambdac - Result: Roughness profile

Low-Pass Filter: Removes roughness, leaves waviness and form - Cutoff wavelength lambdaf (typically 10-20x lambdac) - Result: Waviness profile

Waviness Parameters

Similar to roughness but with longer cutoff:

Wa: Average waviness (analogous to Ra)

Wt: Total waviness height (analogous to Rt)

Typical Causes: - Machine vibration - Spindle runout - Workpiece deflection - Tool chatter - Regenerative vibration

Combined Symbols

Drawing callouts may specify both roughness and waviness:

Example:

Ra 1.6

Wa 0.05

Means: Ra $\leq$ 1.6 μm , Wa $\leq$ 0.05 mm

21.9.3 Contact vs. Non-Contact Methods

Contact Profilometry

Stylus Profilometer:

Components: - Diamond stylus (typically 2-10 μm radius, 60-90° cone angle) - Precision linear stage - Inductive or optical displacement transducer - Signal processing and filtering

Operation: 1. Stylus contacts surface with light force (0.7-4 mN typical) 2. Traverse surface at constant speed (0.5 mm/s typical) 3. Vertical displacement recorded 4. Profile digitized and filtered 5. Parameters calculated

Advantages: - Well-established technology - Traceable to international standards - Works on most materials - Relatively inexpensive - Portable instruments available

Disadvantages: - Contact can scratch soft surfaces - Stylus radius limits resolution - 2D profile only (one line) - Slow measurement - Cannot measure steep walls or re-entrant features

Typical Specifications: - Vertical resolution: 0.01-1 nm - Vertical range: 1-10 mm - Horizontal resolution: 0.01-1 μm - Traverse length: 1-300 mm

Non-Contact Optical Methods

White Light Interferometry:

Principle: Uses interference fringes from white light to measure height

Characteristics: - Vertical resolution: Sub-nanometer - 3D surface maps (not just line profiles) - Very fast measurement (seconds) - Ideal for smooth surfaces

Applications: Polished surfaces, optics, silicon wafers, fine grinding

Laser Triangulation:

Principle: Laser spot imaged by camera at angle; height determined by triangulation

Characteristics: - Non-contact - Fast scanning - Resolution: 0.1-1 μm - Good for rougher surfaces - Handles steep slopes

Applications: Machined surfaces, castings, larger features

Confocal Microscopy:

Principle: Only in-focus light reaches detector

Characteristics: - High lateral and vertical resolution - 3D capability - Handles high slopes - Slower than interferometry

Focus Variation:

Principle: Scans through focus, determines height from sharpest focus

Characteristics: - Works on rough and smooth surfaces - Large vertical range - Handles steep slopes and edges

Comparison

Method	Resolution	Speed	Surface Type	3D Capability	Cost
Stylus	Good (nm)	Slow	Most	2D line	Low-Med
White Light	Excellent (sub-nm)	Fast	Smooth	3D area	High
Laser Triangulation	Moderate (µm)	Fast	Rough	3D area	Med-High
Confocal	Excellent	Moderate	Most	3D area	High

Selection Criteria: - **Stylus:** General purpose, standard compliance, budget-conscious - **Optical:** Soft materials, 3D surface characterization, high-throughput

21.9.4 Effect of Machining Parameters on Surface Finish

Surface finish is directly influenced by machining conditions.

Milling and Turning

Theoretical Surface Roughness (Turning):

$$Ra_{\text{theoretical}} = f^2 / (32 \times r)$$

Where: - f = feed per revolution (mm/rev) - r = tool nose radius (mm)

Example: Feed = 0.2 mm/rev, nose radius = 0.8 mm

$$Ra = (0.2)^2 / (32 \times 0.8) = 0.04 / 25.6 = 1.56 \text{ µm (61 µin)}$$

Milling (similar relationship):

$$Ra_{\text{theoretical}} \approx fz^2 / (8 \times r)$$

Where fz = feed per tooth

Actual vs. Theoretical:

Actual surface finish is worse due to: - Tool wear - Built-up edge (BUE) - Vibration - Material properties - Cutting conditions

Typical: Actual Ra = 2-5x theoretical Ra

Key Factors Affecting Finish

Feed Rate: - Lower feed -> Better finish - Most significant controllable parameter - Quadratic relationship (halving feed reduces Ra by 4x)

Tool Nose Radius: - Larger radius -> Better finish - But increases cutting forces and risk of chatter

Cutting Speed: - Too low: Built-up edge (poor finish) - Optimal range: Good finish - Too high: Excessive wear (poor finish)

Tool Wear: - Fresh tool: Good finish - Moderate wear: Often improved finish (burnishing) - Excessive wear: Poor finish, work hardening

Depth of Cut: - Minimal direct effect on finish - Affects forces and potential deflection

Coolant/Lubrication: - Proper coolant: Reduces BUE, improves finish - Inadequate: Heat, BUE, poor finish

Machine Rigidity: - Rigid setup: Minimal vibration, good finish - Lack of rigidity: Chatter, poor finish

Improving Surface Finish

Machining Strategies:

1. **Reduce feed rate:** Most effective but increases cycle time
2. **Increase nose radius:** Within limits of tool strength and chatter resistance
3. **Optimize cutting speed:** Avoid BUE formation range
4. **Use sharp tools:** Replace worn tools promptly
5. **Proper coolant:** Type and application method
6. **Finishing passes:** Light depth, low feed, high speed
7. **Tool geometry:** Positive rake, sharp edge, honed radius

Advanced Techniques:

High-Speed Machining: - Very high speeds with small chip loads - Excellent finish on aluminum and other materials

Burnishing: - Post-machining rolling operation - Plastic deformation smooths surface - Can achieve $R_a < 0.2 \mu\text{m}$ from R_a 3-6 μm

Vibratory Finishing: - Abrasive media tumbling - Uniform finish on complex parts

Summary

Surface finish measurement quantifies the texture resulting from manufacturing processes. Multiple parameters (R_a , R_z , R_q) characterize different aspects. Contact stylus profilometers are the standard method, while optical methods offer non-contact 3D measurement. Waviness and form must be separated from roughness by proper filtering. Machining parameters—especially feed rate and tool geometry—directly affect surface finish. Understanding these relationships enables optimization of processes to achieve required finish while minimizing cycle time.

Key Takeaways

1. **R_a** is the most common surface roughness parameter worldwide
 2. **R_z** is more sensitive to peaks and valleys than R_a
 3. **Cutoff length** separates roughness from waviness
 4. **Evaluation length** is typically 5x cutoff length
 5. **Stylus profilometers** are standard but contact surface
 6. **Optical methods** enable fast, non-contact 3D measurement
 7. **Feed rate** has the greatest effect on surface finish (quadratic relationship)
 8. **Tool nose radius** affects finish (larger radius = better finish)
 9. **Actual finish** is typically 2-5x worse than theoretical
 10. **Surface finish affects** function, wear, fatigue, and corrosion resistance
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Review Questions

1. What is the difference between R_a and R_z surface roughness parameters?
2. Calculate the theoretical R_a for turning with feed = 0.15 mm/rev and nose radius = 0.4 mm.
3. What is the typical relationship between evaluation length and cutoff length?
4. What are the advantages and disadvantages of stylus profilometers versus optical methods?
5. Why is actual surface finish typically worse than theoretical predictions?
6. How does feed rate affect surface roughness in turning or milling?
7. What causes waviness in machined surfaces, and how is it separated from roughness?
8. List three methods to improve surface finish without changing cutting speed.